

HOW TO STUDY IN COLLEGE

WALTER PAUK • ROSS J. Q. OWENS • TENTH EDITION

gaining



retaining



explaining



How to Study in College

TENTH EDITION

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How to Study in College

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To the Instructor of the Tenth Edition

Students who are seeking help are not primarily interested in theory, and most of them have little patience with merely inspirational talk. They want practical instruction on how to succeed academically. They want something that they can readily understand and apply and that works. After a week of classes, they discover that the hit-or-miss tactics that got them through high school are grossly inadequate and inefficient at the competitive college level. So they turn to us for help.

Let's then teach these students proven techniques for studying and learning. *How to Study in College* is brimming with exciting techniques, based on widely tested educational and learning theory, that have already helped myriad students. But the tail of theory is never allowed to wag the practical, feet-on-the-ground dog. While theory is always implicit and is sometimes given in enough detail to explain the rationale behind a particular technique or reassure the skeptic, it is never presented without explicit applications and never used simply as exhortation. After all, the person who needs penicillin is hardly cured by learning the history of antibiotics!

Because it is so crucial that students learn for the long term, we are wholeheartedly against techniques that stress mere memorization. Such techniques fill the mind with "knowledge" that melts away after a test and leaves learning still to be done. The techniques presented in this book result in real learning. And real learning, like a real diamond, lasts.

Finally, no textbook—no matter how complete or current—is truly useful if it is boring, confusing, or excessively difficult to read. We have worked hard to keep this book well organized and clear, maintaining a conversational tone so that reading it is like having a sincere, person-to-person chat.

What's Different in the Tenth Edition?

Although most of the popular elements from the previous edition have been retained, some of the book's familiar material has been enhanced to make it even better organized, more helpful, and more timely. To provide an extra avenue for framing the book's important ideas and techniques, subsections have been annotated with Skill Tags, Web-inspired labels that highlight the core skills that run as themes throughout the book. In addition, three end-of-chapter vocabulary sections—two brand new and one updated—reinforce the importance of word building as a lifelong pursuit.

New! Skill Tags

Learning assessments can be valuable for both instructors and students in providing a picture of each student's particular approach to study. But beyond the abstract insights they often afford, such assessments rarely offer any sort of tangible action plan. This can be a wasted opportunity, particularly for students, who may appreciate the brief introspection but inevitably forget any revelations because most avenues for follow-up lead nowhere or wind up as dead ends. We've decided to do things a little differently. The updated learning assessment for this new edition (available online at www.cengage.com/success/Pauk/HowToStudy10e) emphasizes eight core skills that serve as the foundation for the book's 14 chapters and points to specific sections in the tenth edition where those skills are used. The core skills are *focusing*, *overviewing*, *reframing*, *summarizing*, *planning*, *questioning*, *clustering*, and *self-direction*. Using a format that almost any Web-savvy reader should recognize, the book's subsections have all been "tagged" with Skill Tags, labels that indicate the skills they stress. Thus, after reflecting on their responses, students can refer to parts or pages of the book for discussions and techniques relating to each highlighted skill.

New! Vocabulary in Action

As with previous editions, the tenth edition continues to emphasize vocabulary development as a lifelong endeavor. Because words are building blocks of thinking, it is essential that students be given a variety of opportunities to add to and strengthen their vocabularies. Unfortunately, not all students have an instant affinity for learning new words. As instructors know all too well, it is often the approach and not the ultimate goal that can lead to indifference and even resistance in some students. It is for this reason that "Vocabulary in Action" section takes three distinct approaches to vocabulary enrichment. "Say What?" is an updated version of the previous edition's "Words in

Context” section that quotes an eclectic array of thinkers and leaders—everyone from J. K. Rowling to Barack Obama—and asks students to use context to pinpoint the meaning for some of the words they use. “Vocab-u-ladder” uses a word puzzle to encourage students to explore the derivations of words and their relationships by linking two seemingly disconnected words together, using a ladder motif that incorporates a series of clues and provides a group of possible words that can be placed on each of the empty rungs. And finally, “Backstory” goes “up close and personal” with key words taken directly from each chapter, spotlighting their fascinating and sometimes surprising histories. The fact that these words have a direct connection to what students have just been reading adds to the relevance and interest of each story.

New! Chapter Checkups

The end-of-chapter questions have been updated with a new name and a greater emphasis on questions that encourage students to reflect on what they have just read and apply the skills they’ve learned in each chapter to the work they are doing elsewhere.

New! Chapter Topics

Although most of the chapter titles in the tenth edition are similar to those in the ninth edition, each chapter now includes an additional chapter topic (e.g., “Concentration” for Chapter 3, “Stay Focused”) that reinforces the chapter’s core topic. Like the Skill Tags that accompany chapter subsections, these chapter topics are designed to make it easier for readers to pinpoint key information in the book and to get to the heart of the skill or technique being emphasized.

Valuable Features Retained

Although the enhancements to the tenth edition make for an even more compelling and effective text, several of the most helpful and popular elements from the previous edition have been retained in this one.

- **Concept Maps.** Concept maps continue to provide a graphical means of summarizing chapter content. Before students read a chapter, the maps supply advance organizers, which, according to David P. Ausubel, make it easier to learn and remember material in the chapter itself. After students have read the chapter, the concept maps provide a bird’s-eye view, showing that chapter’s main concepts with linking lines that establish relationships.

- **Marginal Questions.** The blue marginal questions that make *How to Study in College* distinctive among study skills textbooks remain, as do the outside margins, which not only make room for these printed questions but also make it easier for students to annotate the text on their own.
- **It's Your Q.** Although most of the book's paragraphs are accompanied by marginal questions that demonstrate the Q System, in the tenth edition—as in the previous one—a few questions in each chapter have been replaced by “It's Your Q” boxes that encourage students to formulate and write in their own marginal questions.

Ancillary Materials

This edition is supported by several ancillaries that are designed to reinforce and enrich the basic book:

The **Student and Instructor Companion Websites** located at www.cengage.com/success/Pauk/HowToStudy10e provide additional materials for students and teachers. The Student Companion Website includes “Practice Quizzes,” “Roots and Prefixes Flashcards,” “Words of Wisdom,” and more. The Instructor Companion Website equips instructors with the author's PowerPoint slides, digital Instructor's Resource Manual, and additional support materials for your course.

The downloadable **Instructor's Resource Manual** provides you with quick access to information on preparing your syllabus, extra multiple-choice questions for each chapter, discussion questions, and reproducible masters relating to the concepts found in the text.

PowerPoint Slides for each chapter of the tenth edition are also accessible through the Instructor Companion Website. Use these slides to enhance any classroom presentation.

If you're looking for ways to assess your students, Cengage Learning has additional resources for you to consider. For in-depth information on any of the following items, talk with your sales rep:

- **College Success Factors Index:** This pre- and post-test determines a student's strengths and weaknesses in areas proven to be determinants of college success.
- **CL Assessment and Portfolio Builder:** This personal development tool engages students in self-assessment, critical thinking, and goal setting activities to prepare them for college and the workplace. The code for this item also provides students access to the Career Resource Center.
- **Noel-Levitz College Student Inventory:** *The Retention Management System™ College Student Inventory* (CSI from Noel-Levitz) is an early-alert, early-intervention program that identifies students with tendencies that contribute to dropping out of school. Students can participate in an integrated, campuswide program. Cengage Learning

offers you three assessment options that evaluate students on 19 different scales: Form A (194 items), Form B (100 items), or Form C (74 items). You will receive an online etoken that provides access to Form A, B, or C. Advisors are sent three interpretive reports: The Student's Report, The Advisor/Counselor Report, and The College Summary and Planning Report.

- **The Myers-Briggs Type Indicator® (MBTI®) Instrument¹:** This is the most widely used personality inventory in history—and it is also available for packaging with *How to Study in College*. The standard Form M self-scorable instrument contains 93 items that determine preferences on four scales: Extraversion-Introversion, Sensing-Intuition, Thinking-Feeling, and Judging-Perceiving.

Also, the new edition may be packaged with the **College Success Planner**, a 12-month week-at-a-glance academic planner. The College Success Planner assists students in making the best use of their time both on and off campus, and includes additional reading about key learning strategies and life skills for success in college.

Cengage Learning's TeamUP Faculty Program Consultants is an additional service available. For more than a decade our consultants have helped faculty reach and engage first-year students by offering peer-to-peer consulting on curriculum and assessment, faculty training, and workshops. Our consultants are educators and higher education professionals who provide full-time support helping educators establish and maintain effective student success programs. They are available to help you to establish or improve your student success program and provide training on the implementation of our textbooks and technology. To connect with your TeamUP Faculty Program Consultant, call 1-800-528-8323 or visit www.cengage.com/teamup.

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¹MBTI and Myers-Briggs Type Indicator are registered trademarks of Consulting Psychologists Press, Inc.

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And finally, in this edition, as in the nine previous ones, Walter Pauk offers very special thanks to his students:

“I am eternally grateful to my many students who have taught me much—so that I may pass on a little more to others.”

W. P.
R. J. Q. O.

To the Student

Why waste time reading a book about study when you could be studying instead? If you're like many students, you are both anxious to learn and impatient to succeed. You want a book that provides the answers you need, techniques that can actually use, and, most of all, that doesn't waste your time. We understand this feeling and, over the course of nearly five decades, have written and refined *How to Study in College* with exactly this sort of purpose in mind.

And yet, this is not a simple test-passer, paper-writer, grade-getter kind of book. That sort of quick-fix approach can be very appealing and it may get you over one hump, but unless your skills are solid and your learning is permanent you'll find yourself repeatedly climbing the same hill. *How to Study in College* is organized from top to bottom in a way that should make it easy to learn, use, and remember. It provides you with tips, techniques, and strategies that are supported by research and that you can use again and again, and it is designed with features that reinforce the very lessons you are learning. In short, you'll not only be reading a book that teaches but also one that practices what it preaches!

How This Book Is Organized

What does it take to be a successful student? Imagine the power of being able to answer that basic question in a single, simple, memorable phrase. It might not always make things easier—studying still takes effort—but it would make them clearer and a lot less intimidating.

Becoming a successful student involves building a strong study skills foundation before gaining, retaining, and ultimately explaining information. That's right. That's the seed from which this entire book has grown. Build a base for information and then gain it, retain it, and explain it. All of the book's 14 chapters branch out from this single idea and have been clustered beneath these basic steps.

Part I: Building a Foundation. Even if it's made from sturdy and resilient materials, a tall tower will still topple if it's built upon a rickety base. In the same way, the gaining, retaining, and explaining you do in college must rest on a firm foundation of basic skills. Incidentally, these are skills that will be invaluable at college but should also serve you well at home, in your career, and for the rest of your life. You don't need to be a college student to benefit from knowing how to set goals (**Chapter 1**), control your time and organize your space (**Chapter 2**), stay focused (**Chapter 3**), and handle pressure (**Chapter 4**).

Part II: Gaining Information. How can you possibly know all there is to know? The world is awash with information. It's out there. You just have to go out and get it. But how? First of all, you need to know how to read. If that sounds obvious or even ridiculous, you may need to rethink what you know about reading. The truth is, there's no "one size fits all" method of reading. The way you read should be adapted to *what* you read. That's the topic of **Chapter 5**. Regardless of how you read, the things you read—and the things you hear—are made up of words. The more words you know and the more precisely you understand those words, the more clearly you'll be able to think. That's why **Chapter 6** focuses on strengthening your vocabulary. Even if it were possible to understand and digest every piece of information you're exposed to, it still wouldn't be wise. Time is short and not all information is created equal. Therefore, you need to be able to focus on the valuable information. **Chapter 7** helps you decide which information is worth keeping and which can be discarded. Finally, not all information comes in the form of words. And not all students learn best by using words. **Chapter 8** explores the other channels of communication we use in our efforts at understanding.

Information is everywhere. You need a plan for going out and getting it.

Part III: Retaining Information. When we talk about *retain it*, we're dealing with another important *-ain* word: your brain. As you'll learn in Part III, your brain's working memory only has room for roughly seven items at once. You can't possibly hold on to a semester's worth of information in the working memory of your brain. Not even close. What's more, the things you keep in working memory evaporate, fast. If you can't quickly find a place to store them, they'll be gone in a matter of minutes. Perhaps forever. The *retain it* section of this book will tell you how to hold on to information more or less permanently. Your primary storage areas will be your long-term memory and your notes. In **Chapter 9**, you'll find out more about how the memory works and learn the things you can do and shouldn't do if you want your memories to last. In **Chapter 10**, you'll learn how to retain valuable information by taking notes efficiently and effectively, using a simple, time-honored system that should work in almost any situation. **Chapter 11** shows you how to read those notes, make sense out of them, and then turn them into a part of your permanent knowledge.

Part IV: Explaining Information. In an ideal world, retaining information would be its own reward. You know, learning for learning's sake. But that's not how college works. In college your performance is measured with grades. The grades you receive depend upon how well you can "show what you know," that is, how successfully you can take the information that you've retained during the semester (and often before) and explain it. The most common kind of explaining in college comes in the form of tests and exams. That's the topic of **Chapter 12**. It provides you with systems, tips, and techniques for translating what you've gained into a test score that you can be proud of. **Chapter 13** focuses on another way of explaining—discussions. Depending on the course, class participation or online discussion threads can play an important role in your grade. But more importantly, you may not know how well you understand something until you have had an opportunity to discuss it with others. You'll learn how to prepare for discussions and how to perform in them in a way that benefits both your understanding of your subject and helps your grade. Finally, **Chapter 14** tackles the writing of research papers. Although it may seem like a specialized skill, as this chapter will demonstrate, much of what you'll need to explain in a paper draws on other skills that you've had a chance to develop and refine up to this point.

*You earn your grade
based on your ability to
explain what you've
gained and retained.*

How This Book Is Designed

Everything about *How to Study in College* is focused on helping you to become a successful student. This mission goes beyond the text of the book and includes the elements that make up the book's design. The at-a-glance table of contents, the concept maps, the chapter checkups, and the marginal questions all act as tools to help you gain, retain, and explain the information in this book. In addition, the "Vocabulary in Action" section at the end of each chapter reinforces the idea that building your vocabulary should be a lifetime pursuit and not just a one-shot deal, while the new Skill Tags found at the start of each chapter subsection reinforce that the dozens of strategies, tips, and techniques in this book are all built on a foundation of eight basic skills.

At-a-Glance Table of Contents

How to Study in College includes the sort of detailed table of contents (TOC) you'd expect to find in almost any textbook. It begins on page iii and extends over several pages. But the book includes another "at-a-glance" TOC inside the front cover that lists the book's

four parts and fourteen chapters and briefly explains the purpose of each part. The at-a-glance TOC is a concentrated version of the book that helps you to gain, explain, and retain the book's contents more easily. The psychological effect of being able to see the book's entire contents on a single page can be dramatic and highly beneficial. Suddenly a book of more than 350 pages becomes more manageable and easier to grasp. Instead of operating in isolation, each important idea can be fit into the overall framework of the book's organization.

Chapter Topics

Most chapter textbooks have titles, but that title doesn't always tell you clearly what the chapter's about. That's why each chapter in *How to Study in College* has both a title *and* a *topic*. The topic is a one- to three-word phrase—like Concentration, Comprehension, or Critical Thinking—that cuts to the very heart of the chapter's subject matter, providing a succinct summation of its focus and making it just a little easier for you to stay on track as you progress through the book. Like the Skill Tags that accompany the chapter subsections, the chapter topics are designed to make each chapter's purpose even clearer.

Concept Maps

Textbook chapters aren't whodunits. Surprises can be fun, but there should be no mystery about how textbook chapters are organized or where they are leading. That's why each chapter in *How to Study in College* gives away the plot at the very beginning. It opens with a map that shows you right up front the topics the chapter will be covering and, just as importantly, how they relate to each other.

These maps should prove helpful in gaining the gist of each chapter before you begin reading—especially if you are a visually oriented learner—and make it easier to review and tie things together after you've finished a chapter.

Use them to survey each chapter before you begin reading. Look at the map as a whole and then note the concepts that make up the map. If you haven't noticed already, the map's top concept is the name of the chapter. Now note how the concepts connect. Turn each concept into a question; the topics connected below each concept should provide a broad outline of the answer.

Use them to review the chapter once you've finished it. Flip back to the front of the chapter and look at the map again. By now each topic name should be more meaningful than it was before you started reading the chapter. Test your knowledge of the chapter's concepts by covering the map and then drawing your own version from memory.

Chapter Checkups

The end-of-chapter questions are designed to teach, not test; you'll find no trick questions and no traps to lead you to an incorrect answer. Take each question at face value and answer it to the best of your ability. Use any incorrect answers you give as opportunities to reread the pertinent portion of the chapter. By rereading and rethinking the question and answer, you will greatly strengthen your understanding of the entire concept. Answers to these questions are available online.

Marginal Questions

If you've already flipped through this book, then you have almost certainly noticed something a little unusual about it. Running along the outside of each page in the marginal area that is blank in most books is a series of questions, one for almost every paragraph.

Why questions? Questions are what make learning come alive. They activate inert facts and turn them into vibrant ideas. If questions aren't a key component of your studying, you may not be truly learning. The book's marginal questions can help you in a number of ways. They provide the following:

1. **Concrete examples** of how each textbook paragraph can be seen as the answer to a question. We don't just tell you to do it. We show you how it's done!
2. **Added motivation** for understanding the book's important ideas. Before you read each paragraph, ask yourself the question in the margin and then set a mini-goal of answering it as you read. If you can answer the question, you'll feel encouraged by the knowledge that you're grasping the ideas in your reading.
3. **A ready-to-use recipe** for reviewing each chapter. You can efficiently do a re-cap of the important ideas on each page by covering the text with a sheet of paper, asking yourself the marginal questions, and then checking your responses against the actual text.
4. **Navigational cues** that make it easier to go back and pinpoint particular points in a chapter. If you need to return to a chapter to look up a specific passage or confirm a particular fact, the marginal questions will provide additional signposts that point you to the information you are searching for.
5. **Practice!** Not all of the paragraphs have marginal questions. (See "It's Your Q" below.) That's no oversight; it's done for a reason. With questions above and below you, you'll have examples and context to help you fill in the blanks with your own marginal questions as you work at perfecting this vital part of the learning process.

It's Your Q

The Q System uses marginal questions to encourage active reading. In each chapter, most but not all of the paragraphs are accompanied by marginal questions. At the end of each chapter, the “It’s Your Q” component encourages you to go back and test your ability to come up with your own marginal questions for the chapter paragraphs that don’t already have them. Look for the “It’s Your Q” boxes sprinkled throughout each chapter’s margins and test your ability to apply the Q System yourself by pinpointing the paragraph’s main idea and then writing in a question that will serve as a cue (Q) for that idea.

Say What?

Also at the end of every chapter, you’ll find “Vocabulary in Action,” a three-part section devoted to vocabulary building. “Say What?” features quotations from notable thinkers and leaders. In each quotation, one or more words is italicized and you are asked to select from one of three words that most nearly reflects the meaning of the italicized word. This is not a test; rather, the purpose is to expose you to words in a real-world context. You may select unfamiliar words for further study. Familiar words will provide reinforcement for your existing vocabulary.

Vocab-u-ladder

The second part of “Vocabulary in Action” provides an entertaining way of reinforcing a fundamental truth about words. Rather than operating in isolation, all words are connected. “Vocab-u-ladder” challenges you to use a list of words and a series of clues to connect two seemingly unrelated words.

Backstory

When you learn the histories of words they can acquire a new and deeper significance. That’s the goal of “Backstory,” the third and final part of “Vocabulary in Action.” Backstory takes key words from each chapter and provides the fascinating and sometimes surprising histories of their origins. Once you learn the story behind the word “deadline,” for example, it’s unlikely that you’ll ever view another due date in quite the same way again.

Why do we devote so much of a book about study skills to words? Because when it comes to gaining, retaining, and explaining, nothing is more important. A wide and

precise vocabulary is really the main ingredient or quality that provides all of us with the endless ability for better thinking and judgment, not just in studying but in all phases of life—personally, socially, and professionally.

Skill Tags

There's another important new element included in this book. In addition to the questions that run alongside most of the paragraphs, there are blue and white labels accompanying many of the chapter's subheadings. These are the Skill Tags, the basic building blocks from which the book has been constructed. They alert you to the skill or skills involved in that part of the chapter. For example, reviewing a lecture (see "Relive Your Lectures," p. 268) uses the core skills of *overviewing*, *summarizing*, and *questioning*; changing your time habits (see "Change Your Time Habits," p. 27) uses *self-direction* and *reframing*.

Although the book has four parts, 14 chapters, and well over 100 sections and sub-sections, all the lessons, strategies, tips, and techniques in *How to Study in College* rely on just eight core skills: *focusing*, *overviewing*, *reframing*, *summarizing*, *planning*, *questioning*, *clustering*, and *self-direction*.

1. **Focusing.** Focusing is the ability to zero in on what's important to the exclusion of everything else. This can mean concentration, of course, but it also includes separating main ideas from supporting details and choosing valuable facts from among irrelevant ones.
2. **Overviewing.** Overviewing is a key part of preparation. It involves taking a broad, big-picture look at information to establish the key points and interrelationships that might not be as readily apparent from closer examination. It is the ability to identify the framework or skeleton of information. This creates "advance organizers" that make it easier to arrange and retain more specific information.
3. **Reframing.** While overviewing builds a framework, reframing changes that frame. It is the ability to process the same information from different angles, approaches, or contexts. This can mean adapting quickly to new and changing information or "translating" information from one form, such as words and numbers, into another, such as images. It is also that core skill behind reflection.
4. **Summarizing.** Summarizing means condensing information. Determining the main idea of a paragraph, page, article, or book involves the skill of summarizing. So does reciting, your most powerful defense against forgetting.
5. **Planning.** Planning uses a predetermined system or procedure to approach a particular problem, task, or period of time. Creating lists, budgets, and schedules requires planning, but so does any regular study skills strategy you may use, such as the Cornell System for notetaking.

6. **Questioning.** Questioning epitomizes active learning. It doesn't necessarily involve skepticism—although it certainly can—as much as it entails engaging in a dialogue with information that would otherwise seem inert.
7. **Clustering.** Clustering is the ability to organize or connect things in meaningful patterns. This means sorting and categorizing both ideas and files, for example, but it also includes thinking of reading assignments in terms of phrases, paragraphs, pages, or chapters, depending on which pattern best suits your purposes at the time. It also involves connecting old familiar information to new, unknown information.
8. **Self-direction.** Self-direction is the ability to act on your own, to set your own agenda, and to be the ruler of your own fate. It includes setting goals, cultivating your own interests and motivation, as well as listening to your “inner voice.”

See how you rate with each of these eight fundamental skills. Take the online assessment to determine which skills you excel in and which skills may require additional attention. (Go to www.cengage.com/success/Pauk/HowToStudy10e) With this newfound knowledge, read through the text as you normally would but pay attention to the Skill Tags that accompany the chapter subheadings.

Use your assessment to study more effectively. If, for example, you learn from the assessment that you're good at focusing or relating new information to old but have a difficult time reframing or planning, the knowledge of what your strengths are and where your weaknesses lie can mark the first step to your study success. Skill Tags act as signposts to the skills you need to focus on. They provide an opportunity to sharpen your skills and then use what you know to become a more successful student.

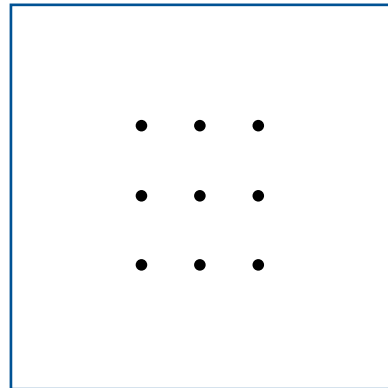
The Value of Reframing

After looking over your answers to the online Skill Tags assessment, you may be thinking, “It's true. I have a hard time finding my old notes, but that's the way I've always been. How can I possibly change?” Or you may believe, for example, that jumping right in and reading a new assignment without surveying it first works better for you because you can save some time and maintain your momentum. These ways of thinking probably feel comfortable, but they may have created blind spots in your view of studying. To get a sense of how blind spots can limit you, try to solve the problem shown in Figure 1. Odds are that a blind spot will prevent you from solving it. Yet once you see the solution, you'll probably say, “How easy! Why didn't I think of that tactic myself?”

The Nine-Dot Problem (Figure 1) not only demonstrates a point, but it is also an excellent learning device. For instance, although very few students have solved the puzzle, they nevertheless have learned the value of reframing. Reframing means remaining flexible enough to look at things from a different angle. If you can break out of the conventional-thinking mold and let your mind rove more freely, you're more likely to be open to more innovative and imaginative approaches to solving problems.

Figure 1**The Nine-Dot Problem**

Connect these dots by drawing four straight lines without taking your pencil from the paper and without re-tracing any lines. The solution appears on page xxiv.



The Power of Planning

It might seem like a big contradiction to suggest that successful students need to be both flexible *and* systematic, but it's true. Being systematic is an important aspect of planning. Without a system, a problem such as the one in Figure 2 can make your head spin.

But if you plan your approach to a problem like this—that is, if you use a system—it can be relatively easy to solve. Drawing a simple grid provides a system for keeping track of the information you know about Anna, Bonita, Carlo, and Derek and the fruits they have brought. Planning can organize information in a way that makes it much clearer. Figure 4 provides the system.

Figure 2**The Lunchtime Problem**

Match each student to the fruit that he or she brought for lunch.

Anna, Bonita, Carlo, and Derek each brought a piece of fruit for lunch. The fruits they brought were an apple, a banana, a grapefruit, and an orange.

1. Bonita's fruit isn't red
2. Anna brought a citrus fruit. So did Derek.
3. Derek's fruit isn't orange

Which student brought each fruit?

Take Advantage of This Book's Additional Resources

Many of this book's important skills and systems are enriched and brought to life on our website. You'll have an opportunity to boost your vocabulary skills with electronic flash cards, to test your knowledge and comprehension with additional questions, and to customize and print out your own copies of the time- and task-based schedules discussed in Chapter 2. You can also learn more academic strategies in online supplementary

chapters on studying foreign languages, studying literature, studying science and mathematics, and using a computer for school. To visit the *How to Study in College* companion website, go to www.cengage.com/success/Pauk/HowToStudy10e.

Take Advantage of Your School's Resources

College or university website. It's a rare college these days that doesn't have a website. What sort of information do these sites contain? That's going to vary widely, depending on the college. Some have simply converted the text of their college catalog into an online form. Others provide elaborate and interactive repositories of information that keep you up-to-date on a variety of college news and often enable you to conduct some transactions that might otherwise need to be done by mail, over the phone, or in person. Regardless of the scope of your college's site, it is often a good place to start. Check the site's FAQs (Frequently Asked Questions) to see if your concern has already been addressed.

College catalog. General information about your college's requirements, policies, programs, and services appears in the college catalog. Even if your college provides this information on its website, it still helps to have a hard copy of the catalog handy during the first weeks of classes to remind yourself of requirements and deadlines to be met.

Student handbook. The student handbook provides information about your school's procedures, regulations, and code of conduct. It may also describe the school's requirements for good academic standing and graduation. For details or for specific department requirements, consult your department office or your academic adviser.

Admissions or registrar's office. You can find answers to questions about grades, transcripts, and college requirements in the admissions or registrar's office. Admission to college and registration for courses begin with this office.

Figure 3
Answer to the Nine-Dot Problem
Begin at the top left corner and follow the arrows.

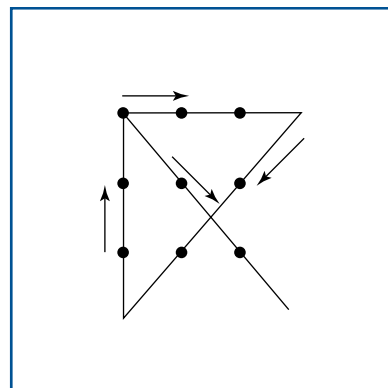


Figure 4

Answer to the Lunch-time Problem

Using a simple grid makes it much easier to sort out the puzzle's clues.

Draw a grid that lists the students in the rows and the fruits in the columns. Mark an $\times$ in a box each time you can conclude that a student could not have brought the fruit to lunch. Put a $\checkmark$ in a box when you can conclude which fruit a student has brought. (Superscripts next to each mark tell us the statement number that allowed us to draw each conclusion.)

	Apple	Banana	Grapefruit	Orange
Anna	$\times^2$	$\times^2$	$\times^3$	$\checkmark^3$
Bonita	$\times^1$	$\checkmark^2$	$\times^2$	$\times^2$
Carlo	$\checkmark^2$	$\times^2$	$\times^2$	$\times^2$
Derek	$\times^2$	$\times^2$	$\checkmark^3$	$\times^3$

1. Bonita's fruit isn't red
2. Anna brought a citrus fruit. So did Derek.
3. Derek's fruit isn't orange

- Statement 1 tells us only that Bonita did not bring the apple.
- Statement 2 tells us that Bonita and Carlo could not have brought either the grapefruit or the orange. Because we already know that she didn't bring the apple, that leaves the banana for Bonita and the apple for Carlo.
- Statement 3 solves the mystery of whether Anna or Derek brought the orange. Because we know that Derek's fruit *isn't* orange, that means he brought the grapefruit and Anna brought the orange.

Office of financial affairs. For answers to questions about scholarships, loans, and grants, contact the financial affairs office. You will come here to pay fees and fines and to pick up your checks if you are in a work-study grant or program. If you want a part-time job on campus for which you must qualify on the basis of your financial status, you will fill out application forms in this office.

Career development and placement office. If you want help choosing a major or setting a career goal, contact the career development and placement office. People in this office can administer various interest, personality, and skills assessment tests to help you determine the kind of work for which you are best suited. They can help you find jobs on and off campus. Some career development centers sponsor on-campus recruitment, inviting businesses to interview prospective graduates and aiding them in submitting applications and résumés. After graduation, you can file a résumé in the placement office if you want your school's help with landing a job.

Academic advising office or counseling department. Academic and guidance counselors can help you with everything from choosing the right course to solving personal problems that prevent you from meeting your academic goals. The academic office or counseling department may be part of the admissions office, or it may be a separate department. In many colleges, students are assigned to an adviser or a counselor who follows their progress throughout their college careers.

Student health center. If you become ill, you can go to a doctor at the health center. The health center may have a pharmacy and may provide a limited amount of hospital care. Some mental health services may be available through this center, through the office of a school psychologist or psychiatrist, or through a peer-counseling group. The health center may also refer students to an agency outside the college.

Student government association. Working with the dean of students, the student government association sponsors student activities such as intramural events, dances, special-interest organizations and clubs, and other social and academic events. (Joining a club or taking part in campus events is a good way to meet other students who share your interests.) In addition, your student government may publish a weekly bulletin or a student handbook that summarizes college requirements and resources.

Student publications. The college newspaper or literary magazine offers contributors unique opportunities for self-expression and provides readers with information and entertainment. Serving on the editorial staff of one of these publications may also fulfill some journalism or English requirements.

Learning lab or skills center. You may turn to the learning lab or skills center for help in improving your study, reading, writing, math, or computer skills. Whether you are required to spend time in a lab because of your performance on a college skills assessment test or you choose to go on your own, take full advantage of the opportunity to gain the skills you need.

Special student services. Veterans, students with physical or learning disabilities, minority students, international students, and students who are economically disadvantaged may need the special assistance of a trained support group to meet their academic goals. If you think you qualify for these services, ask your counselor or adviser about them. Your college may also offer services such as off-campus residence listings.

Athletics office. A listing of the college's athletic programs and events is available in the athletics office. This is the office to visit if you are interested in participating in sports.

Resident assistant. For on-campus students, resident assistants (RAs) can be a great source of information about campus services. Although RAs are not professional counselors, they have recently been through many of the experiences you're undergoing and can probably direct you to the campus office best suited to your needs.

Final Words

Our ultimate goal in this book is to provide you with tools, skills, and systems that will lead to self-sufficiency. Or, as Ralph Waldo Emerson once expressed it: "The best service one person can render another person is to help him help himself."

Skill Tags add a new dimension to study skills

All the lessons, strategies, tips, and techniques in the 14 chapters of this book rely on just eight fundamental skills, which are displayed as Internet-style tags known as Skill Tags throughout this tenth edition of *How to Study in College*:

FOCUSING zeroes in on what's important. Not only is it the key to concentration, but it also lets you separate main ideas from supporting details and choose valuable facts from among irrelevant ones.

OVERVIEWING provides a big-picture look that allows you to identify the framework or skeleton of information. This creates "advance organizers" that make it easier to arrange and retain what you learn.

REFRAMING enables a stronger, more flexible approach to learning. It can mean the ability to adapt quickly to new and changing information or to "translate" information from one form, such as text, into another, such as pictures. It is also the core skill behind reflection.

SUMMARIZING means condensing information. Determining the main idea of a paragraph, page, article, or book involves the skill of summarizing. So does reciting, your most powerful defense against forgetting.

PLANNING uses a predetermined system or procedure to approach a particular problem, task, or period of time. Lists, budgets, and schedules all require planning. So do any regular study skills strategies you use, including the Cornell System for notetaking.

QUESTIONING leads to active learning. By engaging in a dialogue with the things you read, see, or hear, you help bring information to life and transform your notes into knowledge.

CLUSTERING means organizing or connecting things in meaningful patterns, groups, or categories and can involve strengthening newfound facts and information by connecting them to the things you already know.

SELF-DIRECTION is the ability to learn independently and to be the ruler of your own fate. It includes setting goals, cultivating your own interests and motivation, as well as listening to your "inner voice."

Use these skills as versatile study tools

Sharpen these eight skills and you should have the necessary tools to tackle virtually any study challenge – everything from setting goals to writing research papers. Look for these Skill Tags to help determine which tools you'll need.

Take the Skill Tags Assessment to learn where your skills are strongest

Start by finding out how you rate with each of these eight fundamental skills by taking the Skill Tags Assessment at www.cengage.com/success/Pauk/HowToStudy10e. Then, with this newfound knowledge, read through the text as you normally would but pay special attention to the Skill Tags for a better understanding of which skills each section requires. (To learn more about Skill Tags, see "To the Student," page xxi.)

PART

I

Gaining

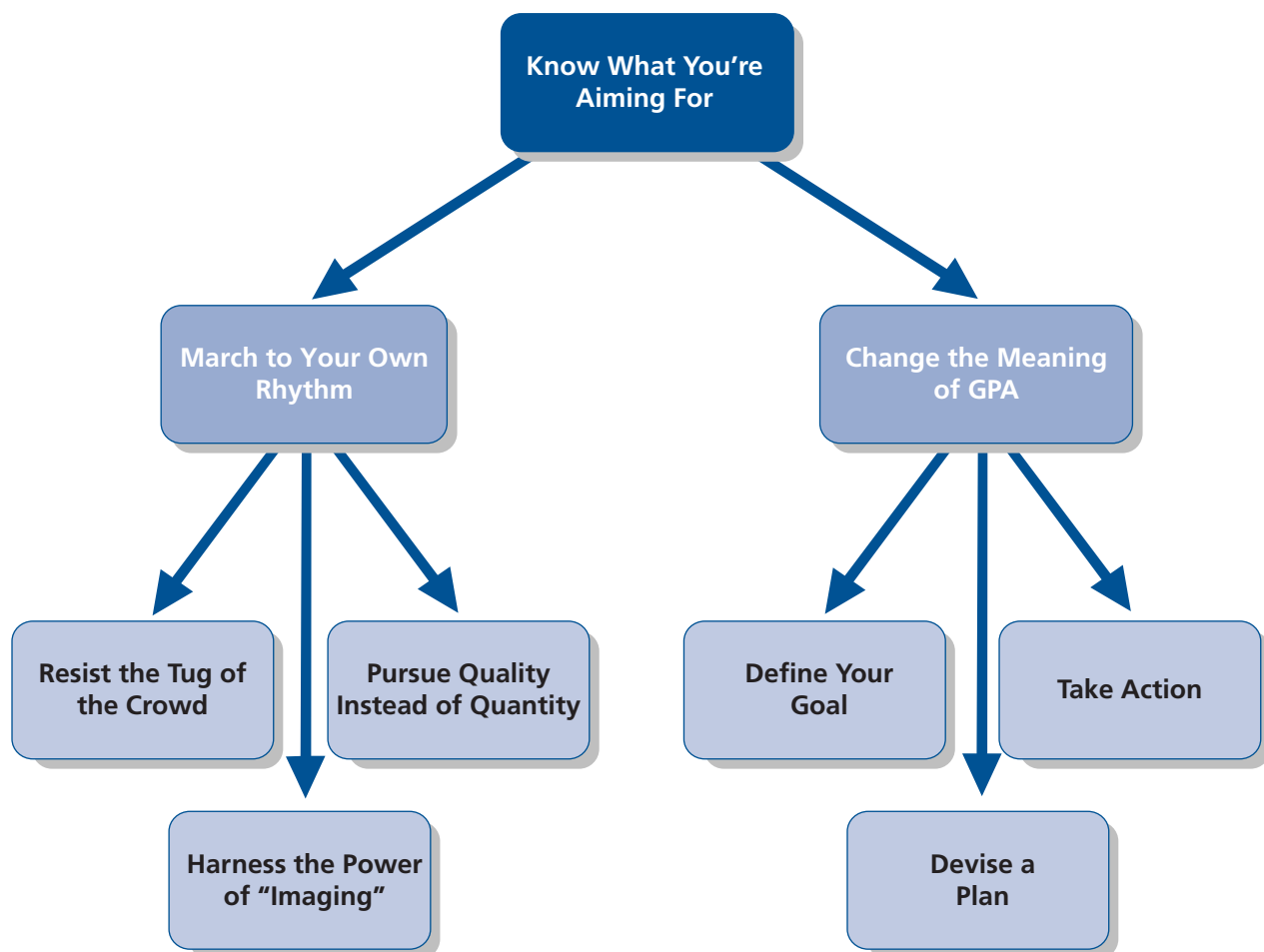
Retaining

Explaining

Building a Foundation

Your success as a student who can effectively gain, retain, and explain information must rest on a strong foundation of skills that will serve you well through college and beyond. Like the basic courses that act as springboards for advance study, learning to set goals, get organized, maintain your concentration, and manage your stress should provide a sturdy base on which you can build your academic success. This part will help you to:

- 1 Know What You're Aiming For**
- 2 Use Your Time and Space Effectively**
- 3 Stay Focused**
- 4 Learn How to Handle Pressure**



Most of us go to our graves with our music still inside us.

Oliver Wendell Holmes
(1809–1890), American physician and writer

Know What You're Aiming For

There's something deep inside each of us that yearns for fulfillment. We were meant for something to which our nature inclines. But what? No one really knows. So, you yourself must dig. Dig—but for what? Start by digging for happiness. Ask, “What kind of work would I be happiest in? Doesn't it make sense that every day's work should be a joy?” In this chapter, you can begin to focus your life as you read how to:

- **March to Your Own Rhythm**
- **Decide What Would Make You Happy**
- **Redefine Your GPA**

What are the risks of having no goal?

Goals bring meaning to life; otherwise, life can be aimless. Viktor Frankl's research revealed these stark data:

When 60 students were asked why they had attempted suicide, 85 percent said the reason had been that "life was meaningless."¹

How do goals go beyond simple survival?

Goals have to go beyond earning your "bread and butter," as is shown in the following slice of human life:

Governments, guided by social scientists, used to say that if you just improve the socioeconomic status of the people, everything will be OK, people will become happy. The truth is that as the struggle for survival has subsided, the question has emerged: survival for what? Ever more people today have the means to life, but no meaning to live for.²

How do goals relate to purpose?

Goals and purpose breathe meaning into life. Goals and purpose form the psychological underpinnings of our individual lives. There's power in purpose, as is shown in this excerpt:

We cannot have deep and enduring satisfaction unless we have self-worth. We cannot have self-worth unless our lives are an earnest attempt to express the finest and most enduring qualities that we are aware of. Purpose is an important condition for an enduring satisfaction with life.³

March to Your Own Rhythm

What was Thoreau's advice?

The nineteenth-century American writer, philosopher, and naturalist Henry David Thoreau famously wrote:

If a man does not keep pace with his companions, perhaps it is because he hears a different drummer. Let him step to the music which he hears, however measured or far away.⁴

How can you bring meaning into your life?

This doesn't mean, as some have suggested, that you need to be an outcast or iconoclast or that you have to live alone in the woods of Walden Pond, as Thoreau once did. Many people who are intricately woven into the fabric of society still manage to adhere to ideals that they have sharpened and defined on their own. To bring meaning into your life, you must decide on your own goal. Avoid the trap of surrendering to the expectations of others instead of pursuing a goal that is personally meaningful.

¹Viktor Frankl, an Austrian psychiatrist and psychotherapist, created the theory of Logotherapy, which states that a person's primary motivation is his or her search for meaning in life.

²Richard J. Leider, *The Power of Purpose* (San Francisco: Berrett-Koehler Publishers, 1997), p. 35.

³Ibid.

⁴Henry David Thoreau, *Walden and Resistance to Civil Government*, 2nd ed. (W. W. Norton, 1992), p. 217.

What role should friends and family play in your goals?

This doesn't mean that you shouldn't discuss your goals with friends and family. It is helpful to ponder their thoughts and advice. But process these through your own brain cells and heart cells, and then make a decision, which must be your very own. Above all, set your own pace instead of marching along with the crowd.

Resist the Tug of the Crowd

REFRAMING

SELF-DIRECTION

What happens when people think as a crowd instead of individually?

When you become part of the crowd, your individual thinking is replaced by crowd psychology. Here's how Gustave LeBon, a French social scientist, described crowd psychology:

The most striking peculiarity presented by a psychological crowd is the following: Whoever be the individuals that compose it, however like or unlike be their mode of life, their occupations, their character, or their intelligence, the fact that they have been transformed into a crowd puts them in possession of a sort of collective mind which makes them feel, think, and act in a manner quite different from that in which each individual of them would feel, think, and act were he in a state of isolation.⁵

What did Gould observe about the psychology of a crowd?

LeBon's observations about crowds were adopted and furthered by Edson Gould, one of the most respected names on Wall Street, who used this easily visualized example to illustrate how the individual is almost powerless to resist the powerful magnetism of the crowd:

You're *alone* in an empty movie theater and hear the cry of "fire." You look around, see no flames, smell no smoke, you *calmly walk* to the nearest exit. But, repeat the same cry of "fire" (again without flames visible or the smell of smoke) in a *crowded* theater and once the *crowd* starts *running* for an exit, you'll find *yourself running*, too. That's crowd psychology.⁶

What should you guard against in crowd psychology?

Don't let crowd psychology rob you of your independence or rob you of your freedom to think and decide individually. In sum, preserve yourself as a sovereign individual so that, for better or for worse, you are the ruler of your career and destiny.

Harness the Power of "Imaging"

REFRAMING

SELF-DIRECTION

In what way does a goal go beyond a statement?

A goal is much more than just a simple statement. Actually, it's a vivid dream that has been mentally acted and reenacted. What lawyer hasn't first imagined presenting a

⁵Quoted in Dan Sullivan, *The Chartist* (December 30, 1997): 4.

⁶Quoted in Edson Gould, *Findings and Forecasts* (New York: Anametrics, 1974).

What is Peale's explanation of imaging?

closing argument to judge and jury? What business executive hasn't imagined outlining an exciting plan to staff members seated around the board of directors' table?

Although *imagined* is generally used to denote imagination, in his landmark book *Positive Imaging* Norman Vincent Peale used the word *imaging* to convey a much deeper concept:

Imaging consists of vividly picturing, in your conscious mind, a desired goal or objective, and holding that image until it sinks into your unconscious mind, where it releases great, untapped energies.⁷

How does imaging your goal affect it?

So, if we think deeply enough and image vividly enough about what we want to do with our lives, our whole being can be energized. Peale, who is one of the most widely read inspirational writers of all time, goes on to say,

There is a powerful and mysterious force in human nature that is capable of bringing about dramatic improvement in our lives. It is a kind of mental engineering. ... In imaging, one does not merely think about a hoped-for goal; one "sees" or visualizes it with tremendous intensity. Imaging is a kind of laser beam of the imagination, a shaft of mental energy in which the desired goal or outcome is pictured so vividly by the conscious mind that the unconscious mind accepts it and is activated by it. This releases powerful internal forces that can bring about astonishing changes in the life of the person who is doing the imaging.⁸

Pursue Quality Instead of Quantity

SELF-DIRECTION

How has money affected the way we define goals?

Over time, Peale's pristine image of a goal has been diluted by the dollar sign. Everyone, it seems, is out to make money. Look at the crowds thronging the racetracks, the casinos, the state lotteries, and even the frequently bubbling stock market. Apparently this "money idea" has steadily seeped into the career goals of students. According to a poll conducted by consulting firm KPMG (Klynveld, Peat, Marwick, Goerdeler), New York, three out of four college students expect to become millionaires.⁹

What advice does John Rau give his students?

The siren song of money may be difficult to ignore, but is it really the key to happiness or the true mark of success? John Rau, who has been the CEO of three corporations, dean of Indiana University's business school, and no stranger to success himself, offers a straight-shooting sanity check for students who are thinking of

⁷Norman Vincent Peale, *Positive Imaging*. Copyright © 1982. Published by Revell, a division of Baker Publishing Group. Used with permission. <http://www.bakerpublishinggroup.com>

⁸Ibid.

⁹Pamela Sebastian, *The Wall Street Journal* (March 25, 1999), p. A1.

What does Dr. Williams conclude about defining your goals?

sacrificing quality in the pursuit of quantity. "Unless you're doing the stuff you like," he cautions, "you can burn out."¹⁰

In line with Rau's succinct pronouncement, Dr. David G. Williams pulls no punches. Few people know more about "burnout" than medical doctors. After years of treating people who were experiencing burnout, Dr. Williams wrote this compelling article on choosing a career that will enhance the possibilities of living a healthier life.

What Do You Want to Be When You Grow Up?

If you're like me, the last time someone asked you this question was quite some time ago. Most of us guys probably responded with something like a fireman, forest ranger, cop, race car driver or even the President. I'm not being sexist (I'm not even sure there was such a word when someone last asked me the question), but if you're female you probably answered the above question with occupations like a model, flight attendant, school teacher or movie star. You chose these answers because at the time you thought these endeavors would be enjoyable. You could visualize being happy. Maybe a more appropriate question than "What do you want to be when you grow up?" would have been "What would make you happy?" And since the answer to this question changes throughout your life, it's a question we really need to ask ourselves more often.

When was the last time you actually got away by yourself and seriously thought about what you needed to be happy? Maybe it's time to do so.

As hectic as life has become, we see happiness as a luxury. Surveys have shown that most people really don't believe it's possible to be happy the majority of the time. They think that true happiness is an unobtainable goal. It's an unpredictable, fleeting sensation over which one has little control. But accepting this idea, that you have no or little control over your own happiness, can have serious health consequences. Happiness is just as important, if not more so, to good health as proper nutrition and adequate exercise. Happiness is a powerful healing force. On the opposite side of the coin we have stress. There seems to be an inverse relationship between stress and happiness. In other words, less happiness leads to more stress. And stress can be an extremely powerful destructive force. Stress is one of the best examples of the power the mind can have over the body.

In animal studies, French researchers at the University of Bordeaux have recently shown that depression and anxiety in adults can be a direct result of placing the mother under stress prior to birth. Stress causes the adrenal glands to produce more of the "stress hormone," corticosterone. Corticosterone easily passes from the mother to the fetus through the placenta. Consistently high levels of corticosterone desensitize brain receptors, altering the feedback system and making it more difficult to shut down the excess

¹⁰Quoted in Hal Lancaster, "Managing Your Career," *The Wall Street Journal* (May 6, 1997), p. B1.

corticosterone production. After birth these receptors in the brain remain desensitized, which can lead to suppression of the immune system, depression and anxiety later in life. (*J Neurosci* 96;15[1 Pt1]:110-6.)

In an amazing study recently performed at Columbia University, New York, researchers found that young girls who suffer from undue stress grow up to be 5 cm. (2 in.) shorter than their happier contemporaries. Stress stunts their growth by depressing the levels of growth hormone in the body.

Volumes have been written on the detrimental effects of stress. And while I won't bore you with all the detailed research here, stress has been linked to everything from asthma and cardiovascular disease to cancer and practically every disease in between. The point to be made here is that happiness replaces and counteracts stress. Probably more than any other single factor, discovering and acting on what makes you happy can improve both the quality and length of your life.

Over the next few hours, days and weeks, I urge you to invest some time in seriously deciding what you want out of life. What would make you happy? I am not talking about what would make you happy for a moment or a day, but instead, what you want and need to be happy for the long term. I can assure you it will be one of the most productive things you will ever do.

Discovering what it would take for you to be happy is, without a doubt, one of the most powerful tools you'll ever possess. It defines your basis for living. It gives you a purpose and provides the answers to life's day-to-day problems. It almost miraculously provides direction at each of life's crossroads. It crystallizes and clarifies your day-to-day goals and activities. It allows you to focus your talents and energies toward achieving the rewards that are most important to you.

If you can't honestly verbalize what you *need* to make you happy, you're going to wander aimlessly throughout life. The clock keeps ticking whether you decide to answer "what would make me happy" or not. Instead of participating and reveling in life, you end up simply reacting to situation after situation. You unquestionably embrace the idea that your own happiness is out of your control. You then begin to believe that it will suddenly appear just as soon as someone or something in your life changes. Unfortunately, that's like playing the state lottery. Your chances of getting hit by lightning are far better than finding real happiness and meaning in your life.

Most of us (I'm as guilty as anyone, if not more so) have a tendency to take life much too seriously. When we were younger it was easier to be less serious. It reminds me of a Bob Dylan song, in which he says, "If you ain't got nothing, you ain't got nothing to lose." The older and more responsible we become, the more we feel we have to lose. We begin to perceive changes in our lives as risks rather than opportunities. As such, we try to avoid change. But in reality, change is not something over which we have any control.

I'm sure you've heard the saying that there are only two things you can count on: death and taxes. Well, there are actually a couple more. One is that your surroundings will change. Technology changes. Weather changes. People change. Everything changes. Always has.

Always will. Accept it. Accept the fact that people and situations are *always* going to be changing throughout your life. Fighting change is like swimming against the current in a river. You're so busy trying to keep your head above water that you never get a chance to see or enjoy what's on the bank. The quicker you accept the fact that everything will change, the quicker you can get out of the water. You can sit on the bank, relax for a moment and evaluate your surroundings. In a life that's always too short, you can then decide how best to spend your remaining time. This brings up the other thing you can always count on—the God-given, human ability to make choices.

Through changes in your thinking, your actions and your lifestyle, you can choose to live your life in a state of unhappiness or in a state of happiness. It's totally up to you.

Although it was several decades ago, I remember sitting at my desk in Mrs. Bengner's first-grade class back in Friona, Texas. Above the chalkboard there were two large handwritten signs. One read, "Act the way you want to be and soon you'll be the way you act." It's probably one of the more lasting lessons I've learned in life thus far. (The other sign said, "One who thinketh by the inch and talketh by the yard should be kicketh by the foot." [I'm still working on that one.])

Before you can "act the way you want to be," and before you can expect to find happiness, you must answer that one simple question. "What would make me happy?" It's a difficult question, probably the most difficult you'll ever have to answer. Getting the answer will require some time and serious thinking. Strangely, there's no right or wrong answer. And even stranger is the fact that only you know the answer. Don't think of this as some kind of test. Nobody is going to give you a grade or set any time limits. The only way you can really fail is aimlessly wandering through life and simply not answering the question at all.

So "What do you want to be when you grow up?" "What would make *you* happy?"

Dr. David G. Williams, "What Do You Want to Be When You Grow Up?" from *Alternatives for the Health Conscious Individual*, Vol. 6, No. 15, September 1996: 119–120. Copyright © 1996, Mountain Home Publishing. Reproduced with permission of Healthy Directions.

Change the Meaning of GPA

What is the best way to become a success?

"If you want to make it in college, your GPA is the key." Students who tell you this are talking about your grade point average, your report card, the number of As and Bs you get in relation to the number of Cs, Ds, and Fs. Grades are certainly important, but they aren't as important as another GPA: your goal, your plan, and the action you take. If you really want to make it, *that's* the GPA you should strive for. If you are able to set a specific goal in your life, if you can come up with an efficient plan for that goal, and

if you have the discipline to take action, there's an excellent chance that you will be headed down the road to success.

Define Your Goal

FOCUSING

What is the purpose of a goal?

Where are you headed? That's the question that your goal is designed to answer. Imagine throwing ingredients into a mixing bowl without any idea of what you are making. Think of running around on the basketball court with no knowledge of the object of the game. The best cooks and the best basketball players know what they are doing and why they are doing it. They have a clear idea of where they are headed. In short, they have a goal in mind.

IT'S YOUR Q

Although winning a basketball game and baking a cake can both be seen as goals, it can be easier to think of your goal as a kind of destination. A lot of our common expressions use this idea. "Making it to the top," "climbing the corporate ladder," and even "reaching for the stars" portray the goal as a place in the distance that you are trying to reach. Of course, some goals really are destinations. When American pioneers declared that their goal was "Pike's Peak or Bust" and tacked signs to their wagons that said so, they were talking about an actual destination hundreds of miles to the west and more than fourteen thousand feet above sea level. When John F. Kennedy made the moon the country's goal in 1961, he was aiming for a destination that was about 238,900 miles out into space. Although Pike's Peak has long since been reached and climbed, climbing a mountain remains a vivid metaphor for reaching one's goal, and although President Kennedy's speech was delivered almost half a century ago, to this day it remains one of the best examples of a clearly defined goal that was achieved through planning and action.

How do minor goals help?

Set Minor Goals

Your life should be full of both major and minor goals. Most of us set minor goals all the time. Passing a test can be seen as a minor goal. So can completing a homework assignment or even finishing a chapter before dinnertime. Having minor goals can be a great help. Each time we reach a minor goal, we get a small sense of victory that helps spur us on toward something even bigger.

Notice in a basketball game how the crowd cheers and the scoring team's pace quickens each time a basket is made. Everyone knows that one basket by itself won't win the game, but when the score is added up, each basket can prove to be crucial. The same is true in school. Although no one has gained success by virtue of a single test or paper, these little victories will add up and help you move toward your major goal. In the meantime, minor goals provide the encouragement you need to cheer yourself on and quicken your pace.

How do you choose your major goal?

Set Major Goals

Choosing a major goal will come naturally for some, while it may be an agonizing process for others. For every person who says, "I've always wanted to be a doctor" or "I

**The Sky's the Limit.
President Kennedy's
ambitious lunar goal
remains a model for
goal-setting to this
day.**

The Goal

*President John F. Kennedy
before a joint session of Congress
May 25, 1961*

Each rocket would send three men into space in order to leave the earth's orbit, to orbit the moon, and eventually to land on the moon and explore it.

The United States sent twenty manned flights into space between May 1961 and June 1969. In July 1969, eight years and two months after President Kennedy set the country's original goal, astronauts Neil Armstrong and Edwin "Buzz" Aldrin set foot on the moon and returned safely to earth.

Your major goal should be large and distant. It should be a target that you can aim for, something to inspire you. Don't let short-term minor goals such as finishing an assignment, passing an exam, and simply getting through the day mark the limits of your dreams. Aim high, but at the same time be sure that the goals you set are specific and distinct. Health, happiness, security, love, and money are all ideals that people aim for, but they are far too vague to be considered goals. On the other hand, "discovering a cure for cancer" and "becoming the best possible parent" are both admirable and specific goals that can help you approach the broad ideals we all share.

What should you do with your goal once you've chosen it?

Make Your Goal Official

If the goal you have chosen is a clear one, you should have no trouble writing it down. Goals that stay only in your head have too great a chance of remaining vague. Furthermore, once you write down your goal, that written record can act as a constant reminder. If you're feeling discouraged, a quick look at your goal can inspire you. (That's what the signs on the covered wagons did.) And if for some reason you forget your goal, a written description can refresh your memory.

What should you do if your priorities change?

The purpose of a goal is not to force you on a course that you don't want to follow; it is to give you a target so that your efforts can be more focused than they would be if you had nothing to aim for. Time and fate have a way of shifting our priorities. People change, and so do the things they view as important. If the goal you once wrote down no longer matches your ambition in life, come up with another goal to replace it.

Devise a Plan

PLANNING

How does your plan relate to your goal?

If you think of your goal as your destination, then a plan can be seen as the route that will take you there. Coming up with a plan is like drawing a map. You need to know where you are starting, where you are heading, and where you plan to stop along the way. Most goals will have several possible plans. The challenge comes in choosing the best one.

How can you choose the most efficient plan?

An efficient plan is a balancing act between what you need and want and what you are able to pay. Paying, as far as a plan is concerned, doesn't always mean money. It can mean time and energy as well. For example, a one-week plan for reviewing your notes is "too expensive" if the test is only two days away. In the same way, a plan that forces you to stay up all night will often cost too much because what you gain in knowledge you will lose in sleep. The most efficient plan will meet your goal without being too costly.

What sort of plans are the best?

The best plans aren't always obvious from the outset. For example, many students approach an exam by answering questions as soon as they receive the test. Given the time limit, that plan may seem to make sense, even though it's actually a bad idea. The most efficient strategy is to read the exam directions, look over all the questions, and even come up with a time plan before answering a single question. The first plan is fast but reckless, whereas the second is steady and dependable. Now you have a more structured and efficient approach to test taking. (For more on this plan, see Chapter 12.) When you make an effort to devise a systematic plan, you will usually gain more benefits than you would with a hastily drawn up strategy.

What sort of impact should flexible thinking have on your goal?

Devising the best plan can require flexible thinking. For example, when you look at a map, you may conclude—as commercial airline navigators once did—that the best way to get from Amsterdam to Tokyo is to head in an easterly direction along what is known as the Mediterranean route. But look at a globe instead of a map, and your perspective

may change. Rather than heading east on the Mediterranean route, commercial planes going from Amsterdam to Tokyo now fly north! That's right. They take what is known as the "polar route," flying over the North Pole to Alaska, and then west to Tokyo—for a savings of roughly fifteen hundred miles! The lesson is this: After you've decided on a goal, work vigorously to accomplish it, but keep looking for ways of achieving the goal more efficiently, perhaps from a different angle. This approach is sometimes known as *reframing*.

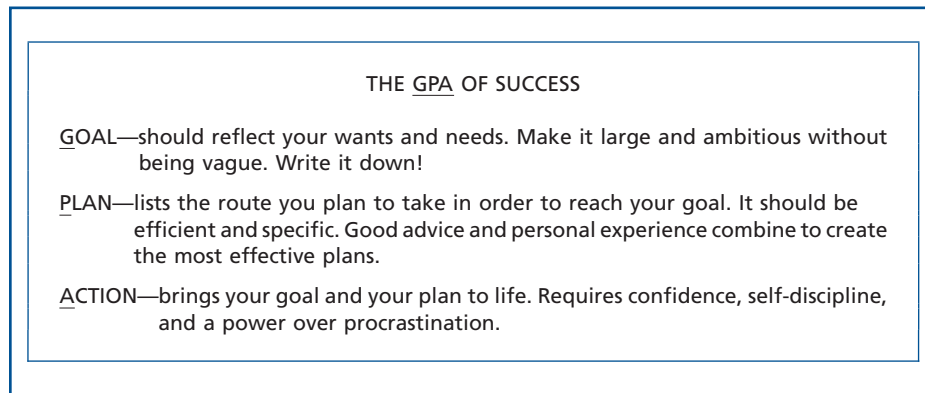
No single plan will work for every goal, and few plans are flexible enough to work for several goals. Using the wrong plan can be inefficient and sometimes even comical. Perhaps you remember the folktale about the lazy son who gets scolded by his mother for losing the money he received as payment from a local farmer. "Next time you get paid," she tells him sternly, "be sure to carry it home in your pocket." But the following day the boy goes to work for a dairy farmer who pays him with a pail of milk instead of money. Anxious not to anger his mother, the boy dutifully pours the milk into his pocket. Although his mother's plan was a good one, it could work only when used in the right circumstance. The same idea applies to your study plans. For example, writing out your notes in full sentences makes sense if the goal is to study a textbook assignment. But if you used the same plan for taking lecture notes, you'd move so slowly that you'd miss most of what the instructor said. The secret is to find a plan that fits the goal you have in mind.

In the same way that good plans may not work for every goal, plans that work for most people may not always work for you. That's why the best way to come up with a plan for success is to balance wise advice with your own experience. This book is full of plans for success and tricks of the trade. All of them have been proven to work, and most should work for you. Use trial and error to determine which plans work best for you, and be prepared to adapt some plans to better fit your needs. Even the best plans can fail if they are used too rigidly. Allow a little breathing room. If things go wrong, don't give up. Adjust and keep on going.

When is a good plan
the wrong plan?

IT'S YOUR Q

Figure 1.2
A New Way to View
G.P.A.



Take Action

SELF-DIRECTION

What is action?

Goals and plans won't do you any good unless you take some action. Action is the spark that brings your goal and plan to life. Without action, goals and plans are moot. You can decide you want to finish a book, and you can even plan the pages that you need to read each day, but until you actually start reading, all your preparations will be pointless. In the same way, the goal to reach the moon and the plans for the spacecraft were impressive, but they didn't come to life until the first rocket left the launch pad and headed into space.

What prevents people from taking action?

Overcome Your Obstacles

Having a goal and a plan is no guarantee that you will take action. Procrastination stops many people from taking action. Procrastination is the tendency to put things off, to write that paper the night before it is due, to cram for a test instead of studying for it right from the start. Although it is just one of many common bad habits, procrastination may be the single greatest obstacle to success. It is also, as we'll see in Chapter 4, a prime source of stress.

How can you prevent procrastination?

The first step in fighting procrastination is to develop a goal and a plan. If you have a goal and a plan but you're still procrastinating, you should take aim at your excuses for not getting your work done. Dream up reasons why you can instead of reasons why you can't. That will often be all it takes to pull yourself out of the vicious circle of inactivity and low self-esteem and put yourself on the road to progress and success.

What does Peale say about taking action?

This chapter would not be complete without more wisdom from Norman Vincent Peale, who expresses the vital importance of *taking action* throughout the entire process of personal goal setting:

I suggest that you write down what you want to do with your life. Until you write a goal, it is only a wish; written, it becomes a focused objective. Put it down on paper. When it is on paper, boil it down to a single sentence: what you want to do, exactly when you intend to start (which should be right now), exactly when you plan to achieve your goal. Nothing fuzzy or hazy. Everything sharp and clear and definite. No reservations or qualifications. Just one strong, simple, declarative sentence I want you to make half a dozen copies of that sentence and put them where you'll see them at least three times a day. I want that pledge to sink down through all the levels of your conscious mind and deep into your unconscious mind, because that is where it will unlock the energies that you will need to achieve your goal.

If setting worthy goals is the first step on the road to success, the second is the belief—the conviction that you are capable of achieving those goals. There has to be in your mind the unshakable image of yourself *succeeding* at the goal you have set yourself. The more vivid this image is, the most obtainable the goal becomes.

Great athletes have always known this. The high jumper “sees” himself skimming over the bar; the place-kicker in football keeps his head down as he kicks, but in his mind's eye

he holds the mental picture of what he wants to happen in the next few seconds The more intensely he images this before it happens, the higher his confidence in himself and the better his chances of making it happen.¹¹

Put Theory into Practice

How can you use your understanding of goal, plan, and action?

Now that you have a clearer understanding of the role of a goal and the way in which a plan and action can turn a dream into reality, you are ready to put theory into practice by writing out your own goal or goals. The following four steps are designed to help you in this process.

Step 1: Brainstorm. On a clean sheet of paper, do some brainstorming about your goals. Jot down possible goals or words about them that come to mind, and do so quickly and freely. Use brainstorming as an opportunity to explore any aspects of any goals you choose. Don't stop to correct your spelling, polish a phrase, reorganize your notes, or analyze a thought. Just keep going until you've jotted down all that you can think of about your possible goals. Next, look over what you've written and group similar items. Formulate each group into a goal by writing a sentence that summarizes its main idea.

Step 2: Plan. Focus in on one of the goals you've arrived at, and write it down as a heading on another clean sheet of paper. Beneath that goal, make a chronological list of all the steps you'll need to take in order to achieve it.

Step 3: List your strengths. On another sheet, jot down those academic and personal strengths that will help you achieve your goal. These could include skills you already have or classes you have taken, as well as personal qualities such as discipline and perseverance.

Step 4: Assess your weaknesses. Identify any academic weaknesses (such as difficulty writing papers) or personal obstacles (such as financial, family, or health problems) that you will have to overcome to reach this goal, and list them either alongside your strengths or on a separate sheet of paper.

What should you do after you've completed the steps?

Don't be surprised if you feel a great sense of relief once you've completed these steps. Even if you're not certain that your goal and your plan are precisely on target, at last you have something concrete that you can adjust and refine. You also have an excellent starting point for guidance and advice. Talk with your academic adviser or with a counselor in your school's career center. Don't underestimate the value of discussing your goals and your plans for achieving them. Get as much feedback as you can. Then, if necessary, modify your goals and plans into realistic, attainable maps for your future. By getting into the goal-setting mode, you can put yourself in control not only of your academic life but also of your life after college.

¹¹From Norman Vincent Peale, *Positive Imaging*. Copyright © 1982. Published by Revell, a division of Baker Publishing Group. Used with permission. <http://www.bakerpublishinggroup.com>

FINAL WORDS

What should you do if you're worried that your goals are going to change?

Goals can change. There's no rule that you can't rethink your objectives after you've defined them. One student seemed genuinely irked that she had to check in with an academic adviser before registering for her first semester of classes. She knew without a doubt that she wanted to go into dentistry and didn't feel that she needed advice from anyone. But the adviser pointed out something she had overlooked. She needed to take an English course in order to meet the college's distribution requirement. That course was a revelation. She fell in love with literature and shifted sharply toward a career in publishing, a goal she pursued with the same passion, determination, and planning that she had originally devoted to dentistry. Yes, goals can change. But that's no excuse to settle for murky, ill-defined objectives. The skills you develop and the lessons you learn in defining your goals will serve you well, no matter where they wind up taking you.

CHAPTER CHECKUP

SENTENCE
COMPLETION

Complete the following sentences with one of the three words listed below each sentence.

1. Based on many of our common expressions, it seems that a goal is considered a _____.
promise **destination** **liability**
2. The best way to come up with a plan is to balance wise advice with your own personal _____.
shortcomings **experience** **success**
3. Goals and plans won't do you any good without _____.
action **education** **discussion**

MATCHING

In each blank space in the left column, write the letter preceding the phrase in the right column that matches the left item best.

- | | |
|-------------------|--|
| _____ 1. Frankl | a. Warned that "burnout" may result if you aren't doing "the stuff you like" |
| _____ 2. Thoreau | b. Defined a clear goal of putting Americans on the moon and returning them safely to Earth |
| _____ 3. LeBon | c. Suggested that we should regularly ask ourselves what would make us happy |
| _____ 4. Gould | d. Suggested that those who seem out of step may just hear a different drummer |
| _____ 5. Peale | e. Noticed the collective mindset that emerges when people become part of a crowd |
| _____ 6. Rau | f. Reported that 85 percent of students who attempted suicide said that "life was meaningless" |
| _____ 7. Williams | g. Used the shout of "fire" in a theater as an illustration of crowd psychology |
| _____ 8. Kennedy | h. Advocated vividly picturing a desired goal or objective in your conscious mind |

TRUE-FALSE

Circle T beside the true statements and F beside the false statements.

1. T F It's not a good idea to discuss your goals with close friends and family.
2. T F When you join a crowd, your individual thinking is replaced by crowd psychology.
3. T F Norman Vincent Peale was a nineteenth-century naturalist and philosopher who lived in the woods near Walden Pond.
4. T F The purpose of a goal is to give you an idea of where you are headed.
5. T F Health, happiness, and security are all excellent career goals.

MULTIPLE CHOICE

Choose the word or phrase that completes each sentence most accurately, and circle the letter that precedes it.

1. When it comes to setting goals, GPA stands for a goal, a plan, and an
 - a. acumen.
 - b. article.
 - c. average.
 - d. action.
2. Passing a test is generally considered to be a
 - a. minor goal.
 - b. major goal.
 - c. common excuse.
 - d. scholastic requirement.
3. Once you arrive at a goal, you should
 - a. stick with it.
 - b. keep it a secret.
 - c. write it down as a reminder.
 - d. come up with a backup goal.
4. Coming up with a plan is like
 - a. running a race.
 - b. building a fire.
 - c. drawing a map.
 - d. none of the above.

5. One of the greatest obstacles to success is
 - a. procrastination.
 - b. lack of money.
 - c. education.
 - d. boredom.

REFLECTION

Think about the ideas outlined in this chapter, and then draw upon your own opinions and experiences to answer each question fully.

1. Most of us think of ourselves both as individuals and as members of at least one larger group. In what ways do you follow Thoreau's advice to march to the beat of a different drummer? In what ways do you adhere to LeBon's observations about crowds?
2. Of the three parts of GPA—a goal, a plan, and action—which is the hardest for you personally? Which is the easiest? Explain.
3. If you haven't done so already, follow the four steps for writing out your own goal. What did you learn about yourself when you did this? Did anything about this process surprise you?

IT'S YOUR Q

The Q System uses marginal questions to encourage active reading. You'll notice that most but not all paragraphs in this chapter are accompanied by marginal questions. Now it's your Q. Scan the chapter for any paragraph that is missing a question, reread the paragraph, establish the main idea, and then arrive at a question that elicits it. Use the questions in the surrounding paragraphs as models for your own marginal questions.

VOCABULARY IN ACTION

The English language is alive. It is constantly growing and changing. The same can be said for your own personal vocabulary. To expand the horizons of your understanding and to refine the precision of your thought, it is vital to continually grow, strengthen, and maintain your own vocabulary. The three exercises that follow are designed to help you with this fascinating, lifelong pursuit.

SAY WHAT?

To gain lasting insight from a quotation, you need a precise understanding of the words it comprises. From the three choices beside each numbered item, select the one that most nearly expresses the meaning of the italicized word in the quote. Make a light check mark (✓) next to your choice.

A good head and a good heart are always a *formidable* combination.

—Nelson Mandela (1918–), former president of South Africa

1. *formidable* impressive frightening polite

Truly successful decision making relies on a balance between *deliberate* and *instinctive* thinking.

—Malcolm Gladwell (1963–), British-born Canadian journalist and author

- | | | | |
|-----------------------|----------|-------------|------------|
| 2. <i>deliberate</i> | educated | intentional | methodical |
| 3. <i>instinctive</i> | habitual | impulsive | primitive |

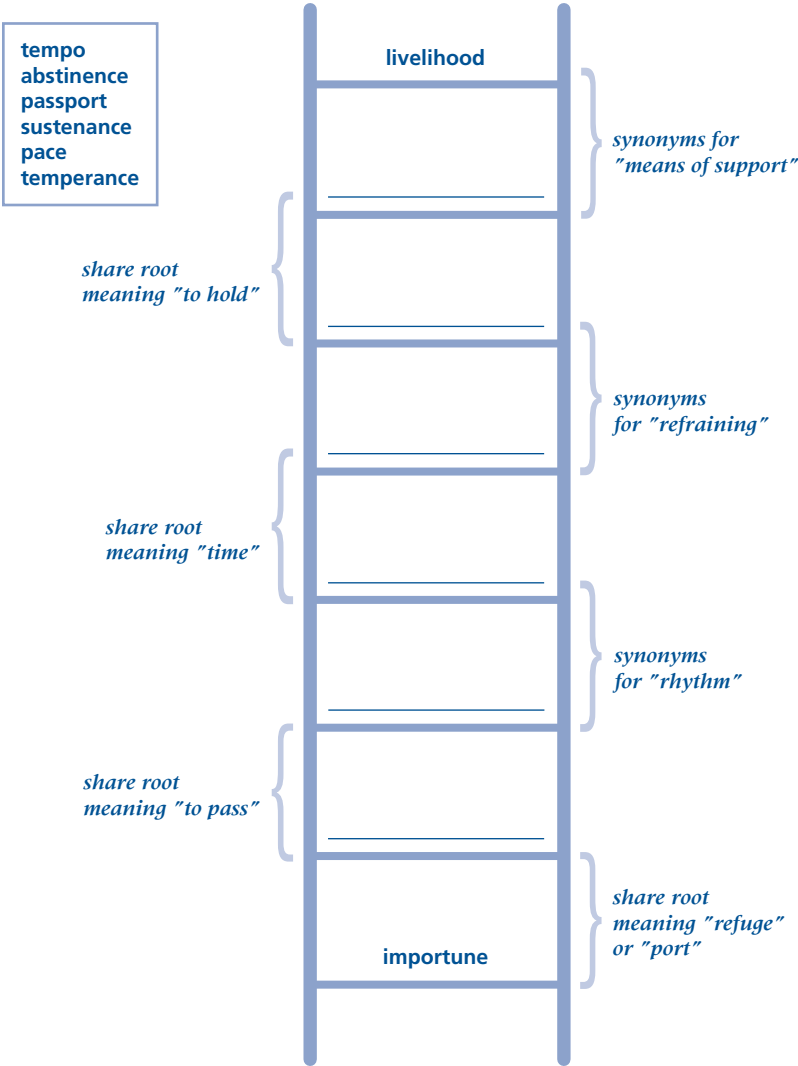
If you do good, people will accuse you of selfish *ulterior motives*. Do good anyway.

—Dr. Robert Schuller (1926–), American evangelist

- | | | | |
|--------------------|------------|---------|--------------|
| 4. <i>ulterior</i> | monetary | hidden | self-serving |
| 5. <i>motives</i> | advantages | profits | reasons |

VOCAB-U-LADDER

Seemingly unrelated words in our language often share numerous connections. Use your knowledge of word synonyms, prefixes, and roots to connect the word at the top rung to the word at the bottom rung, using the words listed below.





Words seldom emerge full-blown into the language. Instead they evolve over time to become the terms we recognize and use today. Each word has a backstory, a sometimes twisted, often entertaining history of how it came to be. Here's the backstory for a word that figures prominently in the chapter you've just read.

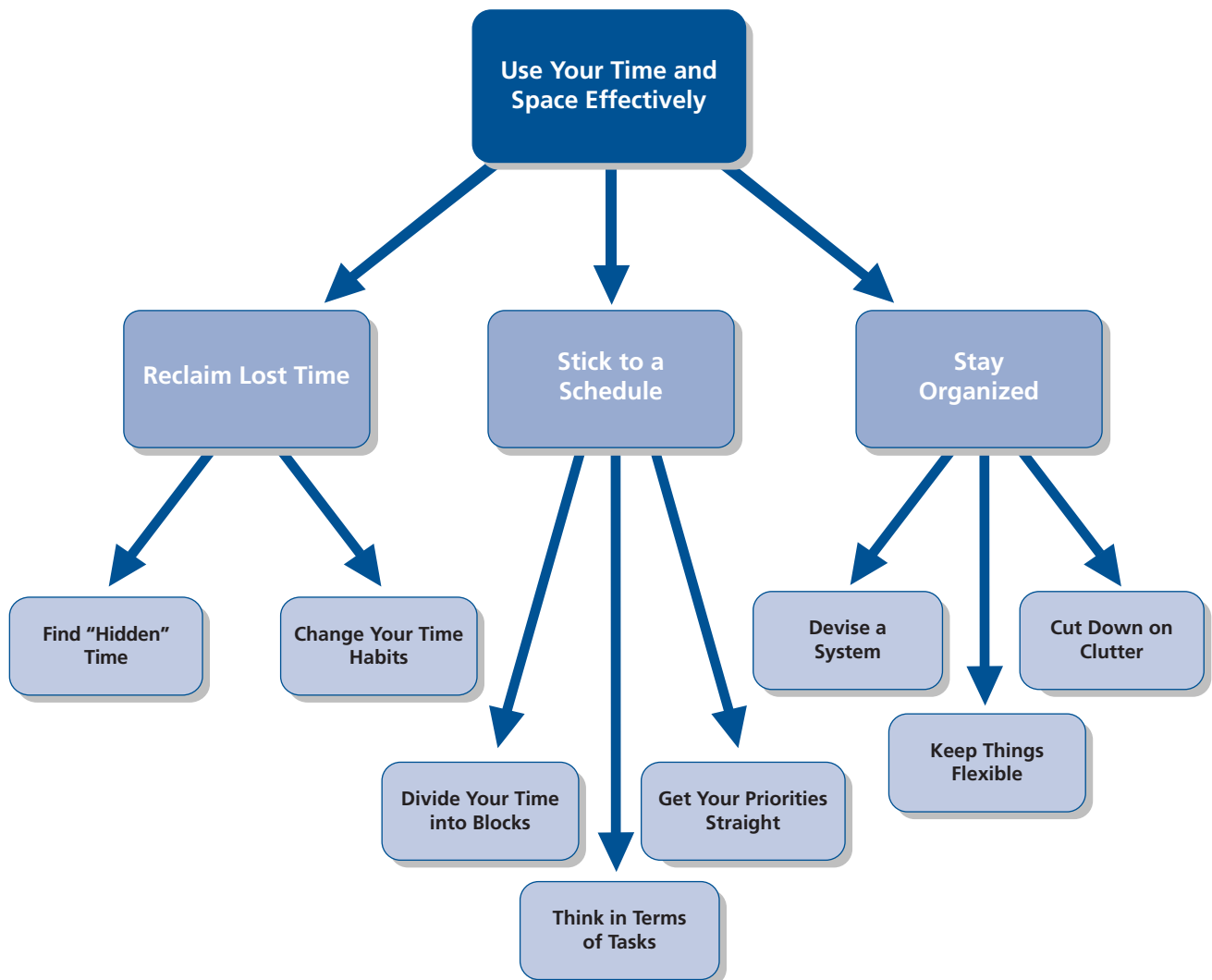
Goal Can you transform obstacles into goals? That's what history did.

goal gōl *n* 1. The purpose toward which an endeavor is directed; an objective. 2. *Sports* a. The finish line of a race. b. A specified structure or zone into or over which players endeavor to advance a ball or puck. c. The score awarded for such an act. 3. *Linguistics* A noun or noun phrase referring to the place to which something moves.*

If you think you know the relationship between *goals*—the things you strive for—and *obstacles*—the things that get in your way—you may be in for a bit of a surprise. Although the history of the word *goal* is not nearly as clear as a good *goal* should be, there's ample evidence that our modern word *goal* originated with the idea of an obstacle. Running as a sport has been with us for centuries. But old race courses weren't as sophisticated as modern tracks. Runners would set off from the same spot and finish when they reached a marker that got in their way. Sometimes it was a mound of dirt. Sometimes it was a pillar. Sometimes it was a fence or wall. What all these finish lines shared in common is that they were all obstacles that not only signaled the end of a race but made it difficult if not impossible to go any farther. The Old English word *gal* meant an obstacle or barrier and the related word *gælan* meant "to hinder." It's out of these words that we get the word *goal*. One person's mound, pillar, or wall is another's indication of victory and accomplishment.**

* Adapted from "goal." *The American Heritage Dictionary of the English Language*, 4th ed. Boston: Houghton Mifflin, 2000. <http://dictionary.reference.com/browse/goal> (accessed May 12, 2009).

** Based on information from "goal." *The American Heritage Dictionary of the English Language*, 4th ed. Boston: Houghton Mifflin, 2000. <http://dictionary.reference.com/browse/goal> (accessed May 12, 2009.); "goal." *Online Etymology Dictionary*. Douglas Harper, Historian. <http://www.etymonline.com/index.php?search=goal> (accessed May 12, 2009.); "goal, *n.*" *Oxford English Dictionary*, 2nd ed. 20 vols. Oxford: Oxford University Press, 1989.; "goal, *n.*" *OED Online*. Oxford University Press. <http://sc-www2.santacruzpl.org:2062/cgi/entry/50096268> (accessed May 12, 2009).



*For every minute spent
in organizing, an hour is
earned.*

Source unknown

Use Your Time and Space Effectively

Time flies, but that's no reason for you to go through each day simply "winging it." Through conscientious use of time and commonsense planning, you can make the most of your day. This chapter ticks off the important elements of effectively managing your time and organizing your space, including how to:

- Find "Hidden" Time
- Change Your Time Habits
- Divide Your Time into Blocks
- Think in Terms of Tasks
- Get Your Priorities Straight
- Stay Organized

How important is time management to your academic success?

Time is a precious and irreplaceable commodity. As she lay on her deathbed, Queen Elizabeth I of England (1533–1603) reportedly said, “All my possessions for a moment of time.” How you use time can determine your success or failure in college. If you use time wisely, you’ll prosper. If you use it poorly, you’ll fail in the job you came to do. That’s why the management of time is the number-one skill to master in college.

How can you gain extra time?

Although many people waste time needlessly and habitually, you needn’t put yourself in the same position. You can gain extra time by reclaiming lost time, by sticking to a schedule, and by staying organized, which will help you use your time more efficiently.

Reclaim Lost Time

How can you put your time to better use?

All of us have claimed that we don’t have enough time to accomplish what we need to do. But the fact is that everyone is allotted the same amount of time: twenty-four hours a day. Many of us allow a lot of this time to go to waste by failing to realize it is available in the first place. In addition, it’s often our day-to-day habits—activities we no longer notice—that save time or waste it. You can put your time to better use by pinpointing areas of “hidden” time and cultivating time-saving habits.

Find “Hidden” Time

SELF-DIRECTION

What is “hidden” time?

There’s a lot of valuable time in your day that is being overlooked simply because you didn’t realize it was time you could use. For those who flush tiny slivers of soap down the drain or toss small scraps of cloth into the wastebasket, there are others who combine those slippery bits and pieces into a whole new bar of soap or stitch discarded shreds into a comfortable quilt. Think of all the time you spend standing in line or even waiting for a traffic light to change. If you could find ways to make use of this “hidden” time, you could almost certainly add hours to each week.

How does pocket work use hidden time?

- *Carry Pocket Work* Many situations may leave you with a few moments of unexpected free time—a long line at the bank or supermarket, a delayed bus or train, a wait at the doctor’s office, a lunch date who arrives late. If you make a point to bring along a book, a photocopied article, index cards on which you’ve written key concepts, vocabulary words, or formulas, you’ll be able to take advantage of otherwise frustrating experiences.

When is your mind free for studying?

- *Use Your Mind When It’s Free* Some activities may afford an overlooked opportunity for studying if you’re prepared. For example, if you’re shaving, combing your hair, or washing dishes, there’s no reason you can’t be studying at the same time. Because

How can recording
information use
hidden time?

many of us tend to “zone out” in such situations, they are excellent opportunities to use time that might otherwise be squandered. Attach small metal or plastic clips near mirrors and on walls at eye level. Place a note card in each clip. Do a math problem or two, or master some new vocabulary words as you eat a sandwich at work.

What is spare-time
thinking?

- *Record Study Information* Another way of using hidden time is by listening to information you’ve recorded on MP3 files or burned onto CDs. Recorded information enables you to keep studying in situations where you’re moving about or your eyes are otherwise occupied, such as when you’re getting dressed or driving. In addition, recorded information can provide a refreshing change from written material.

How can you use your
subconscious to save
time?

- *Employ Spare-Time Thinking* You can make the most of the moments immediately before or after class by recalling the main points from the last lecture as you’re heading to class or by quickly recalling the points of a just-completed lecture as you’re leaving class.

- *Use Your Subconscious* At one point or another, you have awakened during the night with a bright idea or a solution to a problem that you had been thinking about before bedtime. Your subconscious works while your conscious mind is resting. If you want to capture the ideas or solutions produced by your subconscious, write them down as soon as you wake up; otherwise, they’ll be lost. Many creative people know this and keep a pad and pencil near their beds. For example, Nobel Prize winner Albert Szent-Györgyi said, “I go to sleep thinking about my problems all the time, and my brain must continue to think about them when I sleep because I wake up, sometimes in the middle of the night, with answers to questions that have been eluding me all day.”¹

Change Your Time Habits

SELF-DIRECTION

REFRAMING

What’s a good way to
begin saving time?

Habits, by their very nature, are things we do routinely without even thinking. Most of us are unaware of our habits unless someone or something draws attention to them. A good way to take inventory of your time habits is by keeping a daily activities log. From the time you wake up to the time you go to sleep, note all your major activities, the time you started and finished each, and the time each activity consumed. With your day itemized on paper, you can gain a clearer picture of where your time is being spent and where it’s being wasted. The activities log in Figure 2.1 shows one student’s daily routine and how he decided to put his time to better use.

Once you have the concrete evidence of a daily activities log before you, you can begin to eliminate your time-wasting habits and develop or reinforce the time-saving ones.

¹Originally published in *Some Watch While Some Must Sleep*, by William C. Dement, as a volume in The Portable Stanford series published by the Stanford Alumni Association. Copyright © 1972. Reprinted by permission of the author.

Figure 2.1
Record of One Day’s
Activities and
Suggestions for
Making Better Use
of Time



What can you learn from Parkinson’s Law?

Defy Parkinson’s Law Parkinson’s Law says that work expands to fit the time allotted.² To avoid running out of time, work Parkinson’s Law in reverse: For each task, set a deadline that will be a challenge to meet, and strive to meet that deadline. Each time you achieve your goal, reward yourself with some small but pleasant activity. Take a break. Chat with a friend. Stroll around the room. Have a special snack you’ve saved just for the occasion. If you fail to meet a deadline, don’t punish yourself. Just hold back your reward and set another goal. Positive reinforcement is powerful in effecting a change in behavior.

²C. Parkinson, *Parkinson, the Law* (Boston: Houghton Mifflin, 1980).

Why is it so important to obey your alarm?

Obey Your Alarm Clock How many times do you hit the snooze button on your alarm clock before you finally get out of bed? Even one time is too many. Set your alarm for the time you want to get up, not for the time you want to *start* getting up. If you can't obey your alarm, you'll have a hard time sticking to your time schedule. After all, it doesn't even buzz.

IT'S YOUR Q

Limit Texting, E-mail, and Internet Time As marvelous as they can be, text messages, e-mail, as well as the Internet in general can all be tremendous "time sinks," swallowing up hours in a typical day. Rather than checking it constantly, designate specific times during the day when you read and send e-mail or cell phone text messages. It's true that e-mail and texting have sped up communication, but it's a rare message that can't wait a while before being read or sent. The same applies to any Web surfing you may do, whether for schoolwork or pleasure. Time has a tendency to fly by as you click from one link to the next. You can help keep things under control by setting a timer when you surf and returning to your studies when the timer goes off.

What is the value of taking time out?

Take "Time Out" Reward yourself with regular short breaks as you work. Learning in several small sessions, rather than in one continuous stretch, increases comprehension. In one study, students who practiced French vocabulary in three discrete sessions did 35 percent better on an exam than those who tried to learn the words in one sitting.³ So take a breather for ten minutes every hour, or rest for five minutes every half-hour. Whichever method you choose, keep your breaks consistent. This way, you'll study with more energy and look forward to your regular rests. When you return to your desk, you'll find that you feel more refreshed.

How can understanding circadian rhythms affect your scheduling?

Listen to Your Body All of us are subject to circadian rhythms. That is, we have periods when we're most wide-awake and alert and other periods when we're sluggish or drowsy. In general, we're sleepest a few hours before dawn and twelve hours later, in mid-afternoon. In keeping with these natural cycles, we're widest awake about every twelve hours, usually at mid-morning and again in mid-evening. Knowing this can help you plan the day's activities more strategically.

1. *Schedule cerebral tasks for mornings and evenings.* Reading, writing, problem solving, and other "thinking tasks" should be done when you're likely to be most alert.
2. *Save active behavior for mid-afternoon.* Fieldwork, lab work, exercise, and errand running are best done at this time of day, when more sedentary activities may make you feel drowsy. If you're not a heavy coffee drinker, a cup of coffee might get you through the afternoon slump.

³Kristine C. Bloom et al., "Effects of Massed and Distributed Practice on the Learning and Retention of Second-Language Vocabulary," *Journal of Educational Research* 74, no. 4 (March–April 1981): pp. 245–248.

3. *Resist the temptation to sleep in on the weekends.* Changing your sleep schedule on the weekends can have a chain reaction effect on the following week. You may find yourself feeling jet-lagged on Monday or Tuesday if you sleep in on Saturday or Sunday.
4. *Read in the morning; review in the afternoon.* Scientists have discovered that short-term memory peaks at about nine o'clock in the morning and that long-term memory is strongest at about three o'clock in the afternoon.

Stick to a Schedule

What is the function of a time schedule?

A time schedule is a game plan, a written strategy that spells out exactly what you hope to accomplish—during a day, a week, or even the entire term—and how you plan to do it. Committing yourself to planning and keeping to a schedule can seem a bit frightening at first, but following a schedule soon becomes a source of strength and a boon to your life. There are several benefits to a schedule.

How will a schedule provide greater control?

A schedule provides greater control. A thoughtfully constructed time schedule can increase your sense of control in four ways. First, because your schedule is written down, your plans seem more manageable. You can start working without delay. Second, you know you'll study all your subjects—even those you dislike—because you've allotted time for them in your schedule. There's less of a temptation to skip disliked subjects when study time has already been set aside for them. Third, a schedule discourages laziness. You've got a plan right in front of you, and that plan says, "Let's get down to business!" Fourth, you can schedule review sessions right from the start and avoid last-minute cramming for tests.

How does a schedule encourage relaxation?

A schedule encourages relaxation. At the same time, because your plan is written down instead of floating around in your head, your mind is freed for other things. There's no time wasted worrying about what to do next. It's all there on paper. There's no guilt either. Both work and play are written into your schedule. This means that when you take a break, you know you deserve it.

Why are some students reluctant to use time schedules?

Despite these benefits, many students are reluctant to start using time schedules. They feel not only that a schedule will do them little good but also that keeping track of time will turn them into nervous wrecks. Neither worry is warranted.

How does a schedule save you time?

A schedule saves time. Yes, it takes time to devise a schedule, but that time is rewarded. You will be able to shift smoothly from one activity to another, without wondering what to do next.

How does a schedule provide freedom?

A schedule provides freedom. Scheduling frees you from time's control. The people you see dashing from class to library to gym, or eating lunch on the run, are slaves to

time. The students who schedule time, who decide how time will be used, are the masters of time.

In what way does scheduling increase flexibility?
Which schedule type is best: time-based or task-based?

A schedule increases flexibility. Disorganized people often waste so much time that there's no room for flexibility. People who use schedules free their time for a variety of activities and are therefore more flexible.

If you are a full-time student or have considerable control over the hours in your day, your best bet is to rely on traditional schedules that divide your time into manageable blocks. If, however, you are juggling your academic life with the responsibilities of a job or the demands of raising a family, you're probably better off using schedules that focus on tasks instead of on time. Each approach has advantages, but both provide an opportunity to tackle tasks with a genuine game plan instead of flailing at them haphazardly. Regardless of the approach you take, it's important to use your scheduling system not simply as a tool for efficiently filling up your days but as an instrument to help you get your priorities straight.

Divide Your Time into Blocks

FOCUSING

What does the story of the sticks illustrate about time?

A father once tied a bundle of small, thin sticks together with a strand of twine, handed the bundle to his youngest son, and said, "Son, break these sticks in half." The boy used his hands and knees but could not break the bundle. Sadly, he handed it back to his father. Without a word, the father untied the twine, and using only his fingers, snapped each stick one by one.

What is the advantage of dividing your time into blocks?

When the sum total of your obligations and academic assignments seems overwhelming, it helps immensely to split them up into small, manageable units. By dividing each day into blocks, time schedules break up your responsibilities and allow you to deal with them one by one. Assigning a block of time to each activity ensures that you will work at peak efficiency.

What are the components of the three-part scheduling system?

Using time blocks, you can create a three-part scheduling system consisting of (1) a master schedule, (2) a weekly schedule, and (3) a daily schedule. Each plays an integral role in managing your time. The master schedule serves as a basic structure for organizing your activities, the weekly schedule adds specific details to the master schedule, and the daily schedule puts the weekly schedule into a portable form. Although each schedule performs a different function, all three follow the same basic scheduling guidelines.

What is the best way to use big blocks of time?

1. *Don't waste big blocks.* If you have a big block of time, use it for a big assignment. There's a strong tendency to say, "I'm going to get these smaller assignments out of the way before I tackle the big assignment." This is a poor decision. Instead, use the large block of time for a large and time-intensive assignment, and save your small assignments for the little slivers of time.

When it comes to studying, what is prime time?

How does the type of class influence when you should study for it?

Why should you shy away from making your schedule too detailed? What sorts of nonacademic activities belong in your schedule?

2. *Study during prime time.* For most of us, prime time is daytime. In fact, research has shown that each hour used for study during the day is equal to an hour and a half at night. Even so, you may find that you have dead hours during the day when you are less productive than you'd like to be. Schedule less-demanding tasks for these hours.
3. *Study before recitation classes and after lecture classes.* A study session before a recitation or discussion class (a foreign language course or a psychology seminar, for example) helps warm you up. When you walk into class, the material is fresh in your mind. For lecture classes, use the time immediately after class to fill in any gaps in your notes and to review the information you've just learned.
4. *Don't let details tie your hands.* Account for all your time, but don't be overly detailed. The time you'd take to make an overly meticulous schedule can be better used in studying a subject directly, and the chances of your following such a strict plan are slim.
5. *Include nonacademic activities.* Always set aside time for food, sleep, and recreation as well as the other activities of your life. Cheating yourself out of a meal, a good night's sleep, a swim, a family get-together, or a meeting with friends won't save you time in the long run. In fact, it may cost you time because all these activities are necessary for your overall mental and physical well-being. Make your plan for living, not just for studying.

What is the purpose of the master schedule?

What does a master schedule look like?

How does a master schedule help?

Lay a Foundation with a Master Schedule

A master schedule provides an agenda of fixed activities around which your varying activities are arranged. Unless changes occur in your basic program, you need to draw up a master schedule only once per term.

A master schedule grid lists the days of the week at the top and the hours of the day down the left side. The boxes in the grid are filled in with all your required activities: sleep, meals, job, regular meetings, community activities, sports, and, of course, classes. The empty boxes that remain represent your free time. Figure 2.2 provides an example of a typical master schedule.

A master schedule, on a five-by-eight-inch card taped over your desk or carried in your notebook, unclutters your mind. More important, it enables you to visualize actual blocks of time into which you can fit necessary activities.

What is the purpose of a weekly schedule?

Account for Changing Details with a Weekly Schedule

The weekly schedule is built on the foundation of the master schedule. To construct it, photocopy or print out another copy of your master schedule, and fill in the empty blocks with the activities you have planned for the upcoming week. If you have a math test on Friday, for example, you will need to schedule a little extra study time for math. Next week you may be assigned a research paper. If so, you'll probably want to leave

Figure 2.2
A Master Schedule
with Work

	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	Sun.
7-8	← Dress and Breakfast →						
8-9	Bio-Sc		Bio-Sc		Bio-Sc	Dress & Breakfast	
9-10		Yoga		Yoga		Yoga	Dress & Breakfast
10-11	History		History		History		
11-12		Spanish		Spanish		Spanish	
12-1	← Lunch →						
1-2	Math	Graphics Lab	Math	Graphics Lab	Math		
2-3	English		English		English		
3-4		Work-study Prog.		Work-study Prog.			
4-5	Work-study		Work-study		Work-study		
5-6							
6-7	← Dinner →						
7-8							
8-9							
9-10							
10-11							
11-12	← Sleep →						

space in your schedule for library or Internet research. The weekly schedule helps you adapt your time to your changing priorities. Keep it posted by your desk or pasted on the inside cover of your notebook.

A sample weekly schedule is shown in Figure 2.3. The lists that follow show how the guidelines for scheduling were used to set it up.

Monday Through Friday/Saturday

- 7–8 a.m. Avoid the frantic dash and the gobbled (or skipped) breakfast by getting up on time.
- 12–1 p.m. Take a full, leisurely hour for lunch.
- 5–6 Relax before dinner—your reward for a day of conscientious work.
- 7–9 Keep up with current notes and assignments through systematic studying.
- 9–10 To forestall cramming at quiz and examination times, give some time every day to a review of previous assignments and ground covered to date.
- 10 A cease-study time of 10 p.m. provides an incentive for working hard during the day and early evening.
- 10–12 Devote some time every day to reading books that truly interest you. Recreational reading and conversation help you unwind for a good night's sleep.

Tuesday/Thursday/Saturday

- 8–9 a.m. Because chemistry (10–11) is your hardest subject, build your morning study program around it. An hour's study before class will make the class period more meaningful.
- 11–12 p.m. Another hour's study immediately after chemistry class will help you remember the work covered in class and move more readily to the next assignment.

Special

- Tuesday 2–5 p.m. Library: paper
- Sunday 7–9 p.m. English paper
- For some assignments you will need to schedule blocks of time to do research or to develop and follow up on ideas.
- Saturday From noon on, Saturday is left unscheduled—for recreation, for special projects to which you must devote a concentrated period of time, for extra work on difficult subjects, for thorough review.

Figure 2.3
A Detailed Weekly
Schedule Based on a
Master Schedule

Time	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	Sun.
7-8	←		Dress and Breakfast			→	↑
8-9	History	Study Chem.	History	Study Chem.	History	Study Chem.	
9-10	Study History	Swimming	Study History	Swimming	Study History	Swimming	
10-11	Study French	Chem.	Study French	Chem.	Study French	Chem.	
11-12	French	Study Chem.	French	Study Chem.	French	Study Chem.	
12-1	←		Lunch			→	
1-2	Math	Intro to Research	Math	Intro to Research	Math		
2-3	Study Math	Library: Paper	Study Math	↑	Study Math		
3-4	Study English	Library: Paper	Study English	Chem. Lab.	Study English		
4-5	English	Library: Paper	English	↓	English		
5-6	←		Recreation			→	
6-7	←		Dinner			→	
7-8	Study English	Study Math	Study English	Study Math	Study English		English Paper
8-9	Study French	Study History	Study French	Study History	Study French		English Paper
9-10	Review English	Review French	Review History	Review Math	Review Chem.		Study History
10-11	←		Recreational Reading			→	
11-12	←		Conversation, Sleep			→	

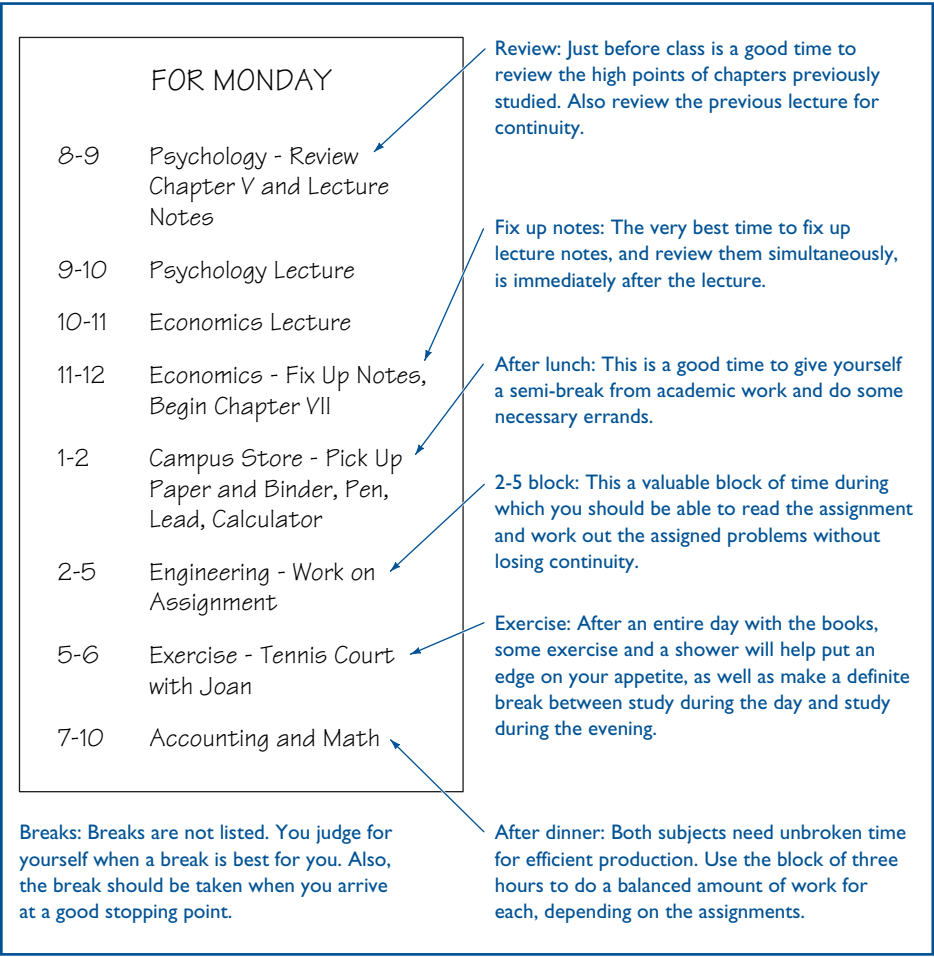
Sunday This is your day until evening. Study history before you go to bed because it is the first class you'll have on Monday morning.

What goes into a daily schedule?

Provide a Portable Game Plan with a Daily Schedule

A daily schedule is a brief yet specific list of the day's tasks and the time blocks you plan to accomplish them in. You should be able to fit all this information on an index card that you can carry around with you all day. Make up your daily schedule each night before you go to bed. Once you have put your worries and concerns on paper, your mind will be free for sleep. You will also have thought through your day and will be better prepared when the morning comes. Figure 2.4 shows one student's daily schedule and explains why it is effective.

Figure 2.4
A Daily Schedule



What are the pros and cons of special scheduling tools?

Use Scheduling Tools If You Feel Comfortable with Them

These days, you can choose from a growing variety of tools and utilities, from computer software to personal digital assistants to Internet applications to hefty loose-leaf notebooks, all designed to make scheduling your time both easier and more intelligent. These tools may provide the breakthrough that some students need to finally begin to appreciate the importance of keeping a schedule. For other students, they are an elaborate, costly, time-intensive distraction that does little to control or organize time. Index cards and blank sheets of paper may not be all that sophisticated, but they are extremely inexpensive, refreshingly flexible, and breathtakingly simple, and they can usually provide all you need to get a firm grasp on your available time. If you swear by your PDA or your store-bought appointment book, that's fine. As long as you keep in mind that your goal is managing time, you can make just about any system work. The essential component is not the tool, but rather the person who is using it.

Think in Terms of Tasks

FOCUSING

When is it preferable to use task-based scheduling?

Because of the way a schedule based on time blocks divides the day into manageable bite-sized segments, it is often preferable for most students. But some of us don't have the luxury of predictably structuring our days from top to bottom. If your days are largely unpredictable or your free time is fluid, you may need schedules that emphasize tasks instead of blocks of time. In addition, long-term assignments, which can't always be squeezed into time blocks, may require a task-based approach.

How did Ivy Lee demonstrate the value of tasks?

Anyone who's ever had the satisfaction of systematically crossing items off a lengthy to-do list has at least an inkling of the effectiveness of a task-based approach. During the first half of the twentieth century, legendary management consultant Ivy Lee demonstrated just how effective—and lucrative—it could be to view things in terms of tasks.

Charles Schwab, then chairman of the Bethlehem Steel Company, challenged Lee: "Show me a way to get more things done with my time, and I'll pay you any fee within reason." Lee thought for a while, then said,

- Every *evening* write down the six most important tasks for the next day in order of priority.
- Every *morning* start working on task #1 and continue until you finish it; then start on task #2, and so on. Do this until quitting time and don't be concerned if you have finished only one or two tasks.
- At the end of each day, tear up the list and start over.

When Schwab asked how much he owed for this advice, Ivy Lee told him to use the plan for a few weeks, then send in a check for whatever he thought it was worth.⁴ Three weeks later, Lee received a check for \$25,000, which is equal to about \$250,000 in today's dollars!

⁴T. W. Engstrom and R. A. Mackensie, *Managing Your Time* (Grand Rapids, MI: Zondervan, 1967).

Figure 2.5
A Task-Based Master
Schedule

	Psychology Research Paper April 21	Train for Amateur Triathlon May 1	Self-Paced Data Mgmt Course
Feb. 7	Select Three Topic Ideas	Up Minimum to 60 Laps	Complete Ch. 1-3
Feb. 10	Do Preliminary Research	Try Ride Up Satyr Hill	
Feb. 14	Make Final Topic Choice	Run 30 Miles Per Week	Complete Ch. 4-6
Feb. 18	Complete Bibliography		
March 15	Finish First Draft		Mid-term Exam
March 18	Begin Rewriting		
April 21	Paper Due		Final

What sort of schedules do you need for a task-based approach?

Of course, just as a daily schedule would provide only a limited view of your tasks, a single to-do list is not sufficient for a task-based approach to help you through the entire semester. Whether it hinges on time blocks or tasks, a three-part scheduling system still makes sense. If you're focusing on tasks, you'll need (1) a task-based master schedule and (2) a weekly assignment-oriented schedule in addition to (3) a daily to-do list.

How does a task-based master schedule differ from a standard master schedule?

Develop a Task-Based Master Schedule

A task-based master schedule enables you to keep track of one or more assignments or goals over an extended period of time. Figure 2.5 provides an example of a task-based schedule. Across the top of the schedule, instead of listing the days of the week, write out the major goals you hope to accomplish or the assignments you plan to complete. Deadlines for subgoals may be written down the left side, where the hours of the day would normally be written in a standard master schedule.

How do you divide your goals into subgoals?

Now divide each goal or long-term assignment into manageable subgoals. List them in a column beneath the task they refer to. For example, if you've been assigned a research paper, you may arrive at the following subgoals: do preliminary research, choose topic, plan outline, conduct research, complete first draft, and revise first draft. As you reach each milestone on the way to completing your assignment, cross it off your schedule. As you do, you provide yourself with visual evidence of and positive feedback for the progress you've made.

When should you use an assignment-oriented weekly schedule?

Make Your Weekly Schedule Assignment-Oriented

If the span of your goal or assignment is a week or less, you can use an assignment-oriented weekly schedule as a supplement to your master schedule. Figure 2.6 shows such a schedule. The format is simple. Draw a horizontal line to divide an $8\frac{1}{2} \times 11$ -inch sheet of paper in half. In the top half, list your subjects, assignments, estimated study times, and due dates. Then, with the due dates and estimated times as control factors, check your master schedule for available time. Allocate enough hours to do the job, and write them on the appropriate line on the bottom half of the sheet. Stick to your schedule. As long as you give study hours top priority, your remaining hours will be truly free.

How should you adjust your daily schedule when your time is unpredictable?

Turn Your Daily Schedule into a To-Do List

Your daily study schedule should simply be a list of things to do arranged in order of priority on an index card. In this case, assigning specific times is likely to lead to frustration.

Figure 2.7 shows a typical daily list. To be successful, you need to develop a habit of referring to your list continually throughout the day. Cross off the tasks as you complete them.

Get Your Priorities Straight

FOCUSING

REFRAMING

When can schedules create a false sense of security?

Even the best schedules can create a false sense of security and accomplishment if they're filled with tasks that may take time but aren't top priorities.

Figure 2.6
A Weekly Schedule
Based on Assignments

Subject	Assignment	Estimated Time	Date Due	Time Due
Electronics	Chapter V - 32 pp. - Read	2 hr.	Mon. 13 th	8:00
English	Paper to Write	18 hr.	Mon. 20 th	9:00
Math	Problems on pp. 110-111	3 hr.	Tues. 14 th	10:00
Industrial Safety	Make Shop Layouts	8 hr.	Fri. 17 th	11:00
Web Development	Generate Website Prototype (6-10 pages)	6 hr.	Fri. 17 th	1:00
Electronics	Chapter VI - 40 pp. - Read	2 1/2 hr.	Weds. 22 nd	8:00

Day	Assignment	Morning	Afternoon	Evening
Sun.	Electronics - Read Chap V English - Find a Topic			7:30-9:30 9:30-10:30
Mon.	English - Gather Notes Math - Problems		2:00-6:00	7:00-10:00
Tues.	English - Gather Notes Industrial Safety	8:00-10:00	3:00-6:00	7:00-10:00
Wed.	English - First Draft Web Development		2:00-6:00	7:00-10:00
Thurs.	Industrial Safety English - Paper Web Development	8:00-10:00	3:00-6:00	7:00-10:00
Fri.	English - Final Copy Electronics		2:00-6:00	7:00-9:30
Sat.				

What’s the difference between efficient time management and effective time management?

Efficient time management is often mistaken for effective time management. Although it’s admirable to work efficiently, that is, to accomplish as much as possible in the time allotted, your efficiency will be wasted if you are focusing on the wrong tasks. As Tom Connellan, an author and consultant, put it, “There’s no point in doing well what you should not be doing at all.”⁵ To manage time effectively, you need to allocate your energy and attention to those tasks that are most important.

⁵Dorothy Lehmkuhl and Dolores Cotter Lamping, *Organizing for the Creative Person* (New York: Crown, 1993), p. 46.

Figure 2.7
A Things-to-Do List

1. Basic Math - 5 problems to solve 2. Geology - Look over specimens 3. Accounting - definitions Bread - 2 Eggs - 1 doz. Margarine - 1 lb.
--

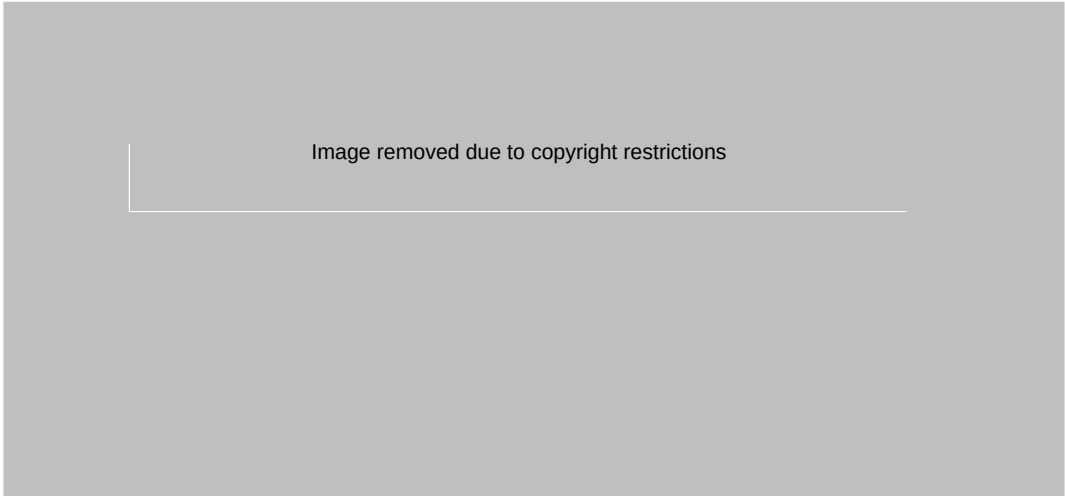
What is the purpose of the urgency-importance matrix?

Why are nonurgent, unimportant tasks considered time wasters?

Well-known author, consultant, and time-management expert Dr. Stephen Covey devised a simple, much-imitated matrix to help you to gain a better understanding of the relative values of the tasks you’re faced with on a regular basis. The matrix has “urgency” on the vertical axis and “importance” on the horizontal one (see Figure 2.8). Tasks that are urgent demand your immediate attention. Important tasks are tasks that will help you to achieve your goal.⁶

The real time wasters are tasks that are neither urgent nor important. They can be easily ignored and do little or nothing to help you achieve either your short-term or long-term goals. These are the sorts of things we often refer to as busywork. If, for example, you are working on the computer but you still take time out to sharpen all your pencils, you’re focusing on a task that isn’t urgent or important. This is clearly the sort of thing that can wait.

Figure 2.8
An Urgency-Importance Matrix
Source: From *Organizing for the Creative Person* by Dorothy Lehmkuhl and Dolores Cotter Lamping. Copyright © 1993 by Dorothy Lehmkuhl and Dolores Cotter Lamping. Reprinted by permission of Random House, Inc.



⁶Ibid., pp. 44–45.

Where do common interruptions fall in the matrix?

Common interruptions—such as a phone call, a text message, or a friend who drops by unexpectedly while you’re studying—tend to fall into the urgent but unimportant category. These things rarely relate to your goals, but they’re hard to ignore. If you need to get work done, it’s generally a good idea to put yourself in a position in which you are less susceptible to these sorts of interruptions. (See Chapter 3.)

How should you handle tasks that are both urgent and important?

Urgent and important items are tasks you can’t ignore and shouldn’t ignore. They include things such as scheduled classes or looming deadlines, as well as more spectacular unscheduled tasks, such as dodging projectiles or escaping burning buildings. If you have a class that starts in five minutes, then that’s urgent and important. You need to attend the course, and you can’t be late. If your paper is due tomorrow, and you’ve hardly started, then you’ve really got no choice but to work until you finish it. All items in this category should get top priority on your task list.

What sorts of tasks are important but not urgent?

The most valuable quadrant in the entire matrix in Figure 2.8 is the one for tasks that are important but aren’t yet urgent. This is where your long-range planning occurs. The more you accomplish in this area, the more successful and organized you’re likely to be. What are some examples of important but not-yet-urgent tasks? Mastering the notes for a recent textbook reading assignment is one. Beginning research on a paper that isn’t due for another month is another. Mastering your notes will put you in a good position the next time a test or quiz is announced (or sprung upon you by surprise). Getting a head start on your paper will prevent you from having to compete with fellow students for limited resources at the library (see Chapter 14) or from having to put in an all-nighter before the paper is due. Some urgent and important tasks (like attending class or leaving the building when a fire alarm sounds) are unavoidable, but, in general, the more time you devote to important but not urgent tasks, the more likely you’ll be able to avoid the tasks that increase your stress, force you to cram, or make you skip sleep.

How should the urgency-importance matrix affect the way you plan out your time?

When planning out your time, tackle the tasks in quadrant 1 first, but see if you can keep them to a minimum, and spend the bulk of your time on tasks in quadrant 2. Avoid adding tasks from quadrant 3 or 4 to your task list. The former are typically going to be unexpected interruptions. Try to establish a study environment where you are less likely to encounter these. And as for the items in quadrant 4, they really don’t belong on any task list. If you find you’re spending time on tasks that are neither urgent nor important, make a point to avoid doing so in the future. Adding such items to your schedule or task list is likely to give you a false sense of accomplishment and distract you from completing truly important tasks.

How can you put the matrix to practical use with task-oriented schedules?

Once you’ve figured out the urgency and importance of each task you plan to accomplish, you can use this information in a number of ways.

1. Go back to a task list you may have already written and number each item depending on whether it’s urgent and important, important but not urgent, and so on.
2. Use the urgency-importance matrix as your actual task list. Looking at which quadrant has more crossed off items will provide a snapshot of how truly productive your day has been.

3. Position your tasks in the matrix first and then create a standard task list, ordering the items by priority based on where you placed them in the matrix.

How can you put the matrix to practical use with a traditional three-part schedule?

If you are using a traditional schedule, the matrix can help you decide how to budget your available time blocks. Begin by using the master schedule as your template, just as you normally would, adding in any tasks or events for the week that already have specific time frames. Then use your matrix to help you decide how to fill in the remaining blocks in your schedule. The weekly view provides excellent reinforcement for those important but not urgent tasks that call for long-term planning.

Stay Organized

What does staying organized have to do with managing time?

Of course, the effort you devote to finding “hidden” time, changing your time habits, and sticking to a schedule will be squandered if the tools you use to tackle your tasks are in a state of disarray. A huge part of being a successful student has to do with being able to find exactly what you need as quickly and efficiently as possible. After all, what good is a carefully planned schedule if you spend the first twenty minutes of the hour you planned to use for mastering your notes, searching for those very notes under a pile of papers or in a maze of nested computer directories? To stay organized and thereby make the best use of that time you’ve so carefully plotted, you need to devise a system, keep it flexible, and cut down on clutter.

Devise a System

CLUSTERING

PLANNING

What sort of organizational system do you need to devise?

Like so many successful study techniques, organizing your work area depends on a system. To a great extent, the kind of system you use doesn’t matter just as long as you have one and it works for you. But in general, it pays to follow the old saying “A place for everything and everything in its place” by designating a logical home base for all the materials you are likely to encounter in your work area. In addition, it makes sense to organize things intelligently, so the materials you need most often are easiest to reach and those you need least are well out of your way. And finally, it helps to logically label items so you can tell at a glance what they contain and where they belong.

Find a Place for Everything

Why is it important to find a place for everything?

There are several reasons why papers, books, and other materials may be obstructing your workspace and even littering your floor, but one of those reasons is so painfully obvious that it’s amazing that it doesn’t occur to those students who have to wade through a sea of supplies each time they try to do some work: *You never found a place for them.* Although some people may be too tired or lazy to put things away, more people

leave things lying about simply because they never figured out where to put them in the first place.

How can you gradually find places for your possessions?

If the prospect of designating a place for everything you own seems complicated and scary, start gradually by designating broad areas for things: bookshelf for your books, drawer for your notebooks, file box for your papers, basket for your mail, jar for your pencils and pens, and so on. Using that as a starting point, start to subdivide your broader areas by, for example, putting this semester's books on the easiest-to-reach shelf and last semester's books on another or by putting science and math books on the first shelf, history and English books on the second. As you refine your system, you'll find it easier to go directly to what you need. But even from the very beginning, you will be in the right ballpark each time you go to look for what you need.

Organize Things Intelligently

How do you organize things intelligently?

Have you ever reached into the refrigerator at breakfast and found that a ketchup bottle, a covered casserole dish, and half an apple are all blocking your access to the carton of milk? Some people have a workspace that is similarly arranged, where the materials they don't need are right there in front of them, and the supplies they do need take some effort to reach. The ketchup, the casserole, and the apple are all things you can use at some point. You just don't need them now. It makes sense to arrange things so they're where you need them when you need them.

How does the CIA system help you arrange your materials?

Just as you arrange your tasks in order of their priority, you should arrange your supplies in a similar fashion. CIA creates a mnemonic to remind you how to arrange things intelligently; that is, they should be arranged according to whether they are current, imminent, or archived.

- *Current.* Current items are the notes and materials you're using right now. They usually belong on your desktop or somewhere close by. They should all be within easy reach. And there's another implied message in this routine: Before you settle down to study, not only should you have the things you need at your fingertips, but also you should get those things you don't need *out of your way*.
- *Imminent.* Imminent items are those you aren't using right now, but plan on using shortly. These are typically the notes and materials from your other classes this semester. In addition, imminent items may include textbooks or materials from previous semesters if they provide background, support, or context for your current semester's work. Imminent items typically sit on a nearby shelf or in a desk drawer.
- *Archived.* Archived items are those you've used in past semesters that may help you this semester or in some future semester but that aren't necessary for your day-to-day work. This would include your notes from previous classes as well as any textbooks from those classes that you've held onto. Because these items don't have to be easy to reach, you can place them on a less accessible shelf in your bookcase, in a file cabinet, or in banker's boxes. If you do put these items in boxes, be very specific in the way

you label them so that you can still find the right book or set of notes if it turns out that you need them.

Choose Logical Labels

What good are labels?

Imagine how difficult it would be to pick out the right book from a shelf if all the books were shelved backward. Although we depend a great deal on labeling, we frequently take it for granted. Labels are designed to make it easier to recognize and find what you need and to organize things systematically, whether by alphabetizing or by some other system.

What should you label?

Label notebooks, individual note pages, and computer files. If you're putting things in a file cabinet, label those folders as well. If you're burning files onto CDs, be sure to label the CDs so that you can quickly differentiate one disk from another. Anything you're likely to sort through, organize, or search for should be labeled. It just takes a second, and it can save you a lot of time. There are lots of good reasons to label your materials and few if any not to do so.

What should the labels say?

How you choose to label your materials is basically up to you as long as you label things consistently. The idea is to be able to find what you need as quickly as possible. That said, here are some suggestions you may want to adopt:

How do you label folders or computer directories for a course?

Label folders or directories according to the course identifier. Rather than labeling a folder or directories with the title of your course (e.g., "Introduction to Macroeconomics"), it's usually better to start with the course identifier (e.g., "ECON102"), which should be easier to locate when you are rifling through a notebook or cabinet or searching through a computer directory. Also, it will mean that all folders from the same subject when alphabetized will wind up in the same basic location.

How should you organize the different kinds of materials for each course?

Use separate labeled folders or dividers for different parts of a class. Keep class notes, lab notes, textbook notes, handouts, and any other course materials in separate labeled sections of your notebook or in their own subfolders in your file cabinet or computer directory.

What sorts of labels do your notes need?

Add enough information to each note page or file so that it can stand alone. It's almost inevitable that your notes will get shuffled at some point. In fact, there are some situations in which you may deliberately want to shuffle them (see Chapter 11). To ensure that your notes don't get easily mixed up, label each note sheet with a course identifier (e.g., "BIO203" or "Biology 203"), the date you took the notes, and the page number. Put the information on an outside corner, so it's easy to flip through and find what you need. With the Cornell System (see Chapter 10), the outside upper corner is probably best because the label is less likely to get in the way of your summary area jottings or marginal cues. To make any notes you've taken on your computer easier to find, name the file after the date (expressed numerically by year, month, and day) and the course identifier. So, for notes you took on February 12, 2010, for a reading in your Biology 203 class, the file name might be 20100212-bio203. If this type of label seems complicated or a little unfriendly to you now, you'll soon get used to it. And you'll discover how much easier it is to quickly find the notes you need.

How can you keep file folders better organized?

Provide an index for any file folders. If you're keeping materials in a file cabinet, create an alphabetical list of the all the folder names and attach it to the cabinet door. That way you'll be able to search for the right folder before you even open the drawer. Also, it should make it easier to tell if you accidentally have two folders serving the same purpose (such as "Lecture Notes" and "Class Notes").

Keep Things Flexible

REFRAMING

CLUSTERING

Why should you keep your organization system flexible?

Perhaps somewhere there's a gigantic landfill devoted to obsolete organizers, to partially completed plans, to filing systems gone awry, and to calendars that were abandoned or ignored. Why do so many of these systems wind up on the trash heap? It's true that in many cases people simply lose interest, but it's just as likely that the systems were far too rigid and demanding. Ultimately, a simple, bare-bones system that works is more powerful than a complex, powerful one that fails. You can keep your organization system going for the long haul if you keep things flexible, by doing things like using loose-leaf paper instead of spiral notebooks and by keeping each class in a separate binder instead of combining them all into one.

IT'S YOUR Q

Use Loose-Leaf Paper

You want any notes you take to be as flexible as possible. Spiral-bound notebooks may seem appealing, but they put your notes into sort of a straitjacket, limiting the things you can readily do with your notes once you've taken them. Of course, you can tear out sheets from a spiral-bound notebook, but that leaves a residual of paper bits that can sometimes have a negative psychological effect on your productivity. Lined-paper tablets are also popular with some students, and although these tablets aren't quite as messy and limiting as spiral notebooks, they still force you to go through your notes in order. If you do write on lined-paper tablets, buy the three-hole kind and then carefully tear off the pages of notes once you've taken them and store them in a loose-leaf binder. As you'll discover in Chapter 11, being able to separate your notes into individual sheets instead of keeping them as part of a notebook or pad will prove to be a real advantage when it comes time to prepare for a test or quiz. Also, if you keep blank sheets of loose-leaf paper in each notebook, you'll always be prepared, even if you make the mistake of bringing the wrong notebook to class.

Why should you have a different notebook for each class?

Limit Each Notebook to One Class

Standard loose-leaf notebooks can be large and unwieldy, not to mention kind of expensive. Luckily, you don't really require anything so elaborate for your class. All you really need are three rings and a cover. Full-sized, soft-sided notebooks still hold standard-sized paper. But these are thinner and less expensive and enable you to use a different notebook for each class instead of putting the materials from several, often unrelated classes in a single binder. Then you can take only what you need when

you head to class. Why do you need your economics notes when you're in class taking notes for sociology? Besides, most of the loose-leaf additions that improve the organization of a large notebook—pockets, dividers, graph paper, and so on—work just as well in a thinner, more flexible notebook. In fact, thinner notebooks often come with a pocket on the inside of the front and back covers. Also, you can buy a notebook of a different color for each class, making it easy to distinguish one from another.

Cut Down on Clutter

CLUSTERING

FOCUSING

What causes clutter and how can you reduce it?

Although getting organized involves far more than just minimizing clutter, clutter is generally what most people think of when they imagine disorganization. The cause of clutter is pretty straightforward. It usually results when too many things are out of place. If you reduce your level of “stuff” and make a routine of putting the things that remain back in their place, you should go a long way to reducing clutter.

What does the OHIO rule suggest?

Use the OHIO Rule for Papers and E-mail

The acronym OHIO stands for “Only Handle It Once.”⁷ When you get a new piece of paper, do something with it right away. Read it and then throw it out, file it, or if it's something that requires your immediate attention, act on it. Many people needlessly pick up the same sheets of paper again and again without doing anything decisive. That wastes time and it usually leads to clutter. By the way, the same advice applies to e-mail. If your inbox is filled with e-mail messages, you're creating electronic clutter. Delete the message, file it in a directory, or act on it right away if it's urgent. But don't leave it sitting in your inbox, to be scrolled past day in and day out or repeatedly opened and closed.

Why is it a good idea to avoid printing out materials?

Don't Print It Out If You Can Avoid It

One simple way to reduce the paper-based clutter on your desk or work area is to limit the things you print out. In many colleges, your syllabus, your assignments, and other course materials, sometimes including tests and texts, come to you via e-mail or through a website or a course management system. Some students go through “paper withdrawal” and have a tough time keeping all their materials on the computer instead of in a stack of sheets on their desk. There may still be some materials that you prefer to have on paper, but don't be like the person who salts her food without tasting it first. Try to refrain from automatically printing out your materials. Of course, if your course materials are only available in hard copy, you have no choice.

⁷Edward M. Hallowell, “Overloaded Circuits: Why Smart People Underperform,” *Harvard Business Review* (January 2005): 60.

What's the last thing you should do at the end of the study day?

IT'S YOUR Q

What if your desktop is a computer desktop?

What's the advantage of setting out your materials in advance?

Allow Time for Cleanup and Setup

When you reach the end of your study day, instead of abruptly getting up from your chair and moving on to something else, take just a few minutes to stow away the things you've been working on in their proper places and to set out the things you plan to be working on the next time you sit down.

Anything you leave lying around from the previous day is likely to interfere with the work you do the following day. Also, things you leave out have a tendency to pile up. After all, no one plans to have a messy desk or work area. It tends to happen over time. Most cluttered surfaces are covered with items that were left there just "temporarily." Take a few minutes to put books, notebooks, and other supplies back in their places. The few minutes you devote to this end-of-study ritual will ensure that your well-organized work area remains organized. If you feel as though you're too tired, or you're concerned that it will take too much time, try an experiment: Look at your watch or a stopwatch as you go through the process of logging off your computer, reshelf books, and putting away pens, pencils, and papers. You are almost certain to find that the routine doesn't take long at all. And yet it can save you a good deal of time in the long run.

The practice of straightening your desktop is still helpful, even if that desktop is a computer desktop. In fact, it's usually a lot easier to straighten your computer desktop. The process of moving, storing, and deleting files is much faster when those files are computer files.

If you gather together materials that you plan to use the next day, you'll have an important head start. If you have a class the first thing in the morning and there's a risk that you may have to head out the door a little faster than you'd like, the few minutes you spent the night before, calmly getting everything together, should minimize the risk that you will forget something if you're in a hurry. Also, setting out the next day's materials has an important psychological effect. It gets you to think about the following day's activities and helps to put you in the proper mindset. Finally, assembling tomorrow's materials in advance helps to counteract inertia and discourage procrastination. When you wake up the next day and find that everything you need is there waiting for you, you should find it easier to hit the ground running instead of just spinning your wheels.

FINAL WORDS

Why do people who understand the importance of time continue to waste it?

Let's face it. Most people understand the supreme importance of time and the value of sticking to a schedule, yet many of us continue to waste time needlessly. Why? Perhaps it is because we are human beings, not machines. Learning to use time wisely after years of wasting it doesn't happen by simply flipping a switch. Controlling your time is

ultimately a matter of developing the right mindset. If the dying words of Queen Elizabeth and the compelling evidence of this chapter haven't been enough to convince you to change your time-wasting ways, perhaps the following tidbits about time will finally make things click.

Seven Valuable Tidbits About Time

1. One of America's greatest composers, jazz giant Duke Ellington, still wasn't ashamed to admit, "Without a deadline, I can't finish nothin'."
2. Be a contrarian. Go to the library when almost nobody is there. Get into the dining-hall line before the crowd. Get the reserved books before the line forms. The time you save will add up quickly.
3. Make decisions wisely by asking, "What are the alternatives?" Make a list of the alternatives, and then put pluses and minuses alongside them. Learn this process. It will save lots of time.
4. Don't spread yourself thin by attempting to become an "information junkie." This scattershot approach takes up a great deal of time and can still leave you feeling stressed and dissatisfied. Just make sure that you gain a firm grip on your own field.
5. When you're really through studying, spend an extra fifteen minutes studying just a little bit more.
6. If the thought of saving time sounds sensible but uninspiring, ask yourself this simple question: What do I want to save time for? Suddenly, the efficient use of time may take on a significance that it never had before.
7. A Sanskrit proverb puts everything in proper perspective:

Today well lived
Makes every yesterday a dream of happiness
And every tomorrow a vision of hope
Look well therefore to this day.

CHAPTER CHECKUP

SENTENCE
COMPLETION

Complete the following sentences with one of the three words listed below each sentence.

1. A time schedule functions as a _____.
reward game plan punishment
2. When it comes to studying, “prime time” is usually _____.
daytime mid-afternoon flexible
3. A master schedule should normally be drawn up once a _____.
day week term

MATCHING

In each blank space in the left column, write the letter preceding the phrase in the right column that matches the left item best.

- | | |
|--------------------------|--|
| _____ 1. Pocket work | a. Work expands to fit the time allotted |
| _____ 2. E-mail | b. Method for minimizing paper clutter |
| _____ 3. Master schedule | c. Nobel Prize winner who used sleep time to solve problems |
| _____ 4. OHIO | d. Provides a basic structure for organizing the term's activities |
| _____ 5. CIA | e. Rhythms that influence your body's cycle of sleeping and waking |
| _____ 6. Szent-Györgyi | f. Productivity boon that can sometimes be a “time sink” |
| _____ 7. Parkinson's Law | g. Supplies study material for unexpected free time |
| _____ 8. Circadian | h. Method for arranging your study materials |

TRUE-FALSE

Circle T beside the true statements and F beside the false statements.

1. T F Taking breaks has a detrimental effect on comprehension.
2. T F Schedules don't have to be effective as long as they're efficient.
3. T F Using a time schedule can make you a slave to time.
4. T F Scheduling saves time that might otherwise be wasted.
5. T F An hour of daytime work is usually more productive than an hour at night.

MULTIPLE CHOICE

Choose the word or phrase that completes each sentence most accurately, and circle the letter that precedes it.

1. When your obligations and assignments begin to seem overwhelming, it helps to
 - a. take a nap.
 - b. divide your time into blocks.
 - c. use Parkinson's Law.
 - d. all of the above.
2. "Hidden" time is time that you
 - a. aren't allowed to use.
 - b. managed to overlook.
 - c. reserve for recreation.
 - d. leave off your schedule.
3. The most *valuable* quadrant of the urgency-importance matrix is
 - a. urgent but not important.
 - b. important but not urgent.
 - c. not urgent and not important.
 - d. none of the above.
4. The master schedule provides
 - a. an alternative to a weekly schedule.
 - b. an excuse for increased recreation.
 - c. a basic structure for organizing your activities.
 - d. a solution to the problem of hidden time.
5. A task-based schedule may be your best bet if
 - a. your days are unpredictable.
 - b. your free time is fluid.
 - c. you have a full-time job.
 - d. all of the above.

REFLECTION

Think about the ideas outlined in this chapter and then draw upon your own opinions and experiences to answer each question fully.

- 1. Identify at least two instances of “hidden” time in your typical week and suggest some ideas you’ve arrived at after reading this chapter for putting that time to better use.
- 2. Explain which type of schedule you think will be the most effective for your particular needs and why.
- 3. Use the urgency-importance matrix (see Figure 2.8) to arrange your own tasks in the four quadrants. Which of your quadrants contains the most tasks? Which has the fewest? Were you surprised by the results? If so, why? If not, why not?

IT’S YOUR Q

The Q System uses marginal questions to encourage active reading. You’ll notice that most but not all paragraphs in this chapter are accompanied by marginal questions. Now it’s your Q. Scan the chapter for any paragraph that is missing a question, reread the paragraph, establish the main idea, and then arrive at a question that elicits it. Use the questions in the surrounding paragraphs as models for your own marginal questions.

VOCABULARY IN ACTION

To expand the horizons of your understanding and to refine the precision of your thought, the three exercises that follow are designed to help you grow, strengthen, and maintain your own vocabulary.

SAY WHAT?

From the three choices beside each numbered item, select the one that most nearly expresses the meaning of the italicized word in the quote. Make a light check mark (✓) next to your choice.

The most *erroneous* stories are those we think we know best – and therefore never *scrutinize* or question.

—Stephen Jay Gould (1941–2002), American paleontologist and evolutionary biologist

- | | | | |
|----------------------|-----------|------------|------------|
| 1. <i>erroneous</i> | misguided | inaccurate | misleading |
| 2. <i>scrutinize</i> | inspect | see | criticize |

Chapter 2: Use Your Time and Space Effectively

53

Any good piece of material like Shakespeare ought to be open to *reinterpretation*.

—Denzel Washington (1954–), American actor and director

3. *reinterpretation* translation rethinking duplication

Gentility is what is left over from rich ancestors after the money is gone.

—John Ciardi (1916–1986), American poet and critic

4. *gentility* refinement large debts large family

Those who make the worst use of their time are the first to complain of its *brevity*.

—Jean de La Bruyère (1645–1696), French writer and moralist

5. *brevity* briefness dullness wastefulness

VOCAB-U-LADDER

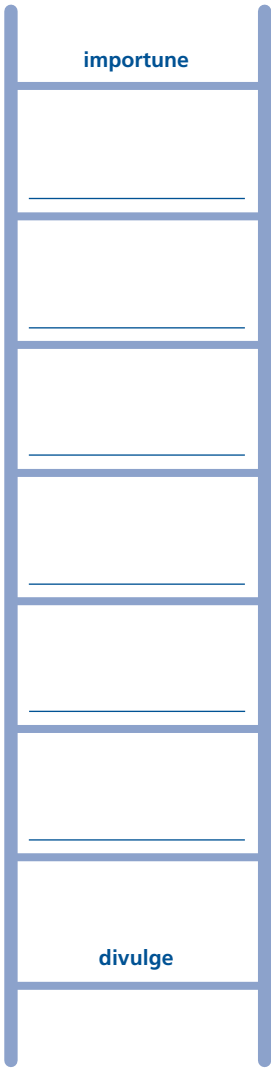
Use your knowledge of word synonyms and roots to connect the word at the top rung to the word at the bottom rung, using the words listed below.

indisposition
coherence
insist
expose
consistency
hesitancy

share root
meaning "to stand"

share root
meaning "to cling"

share root
meaning
"to put or place"



synonyms
meaning "beg"

synonyms for
"agreement"

synonyms for
"reluctance"

synonyms for
"reveal"



Here's the story behind a word that figures prominently in the chapter you've just read.

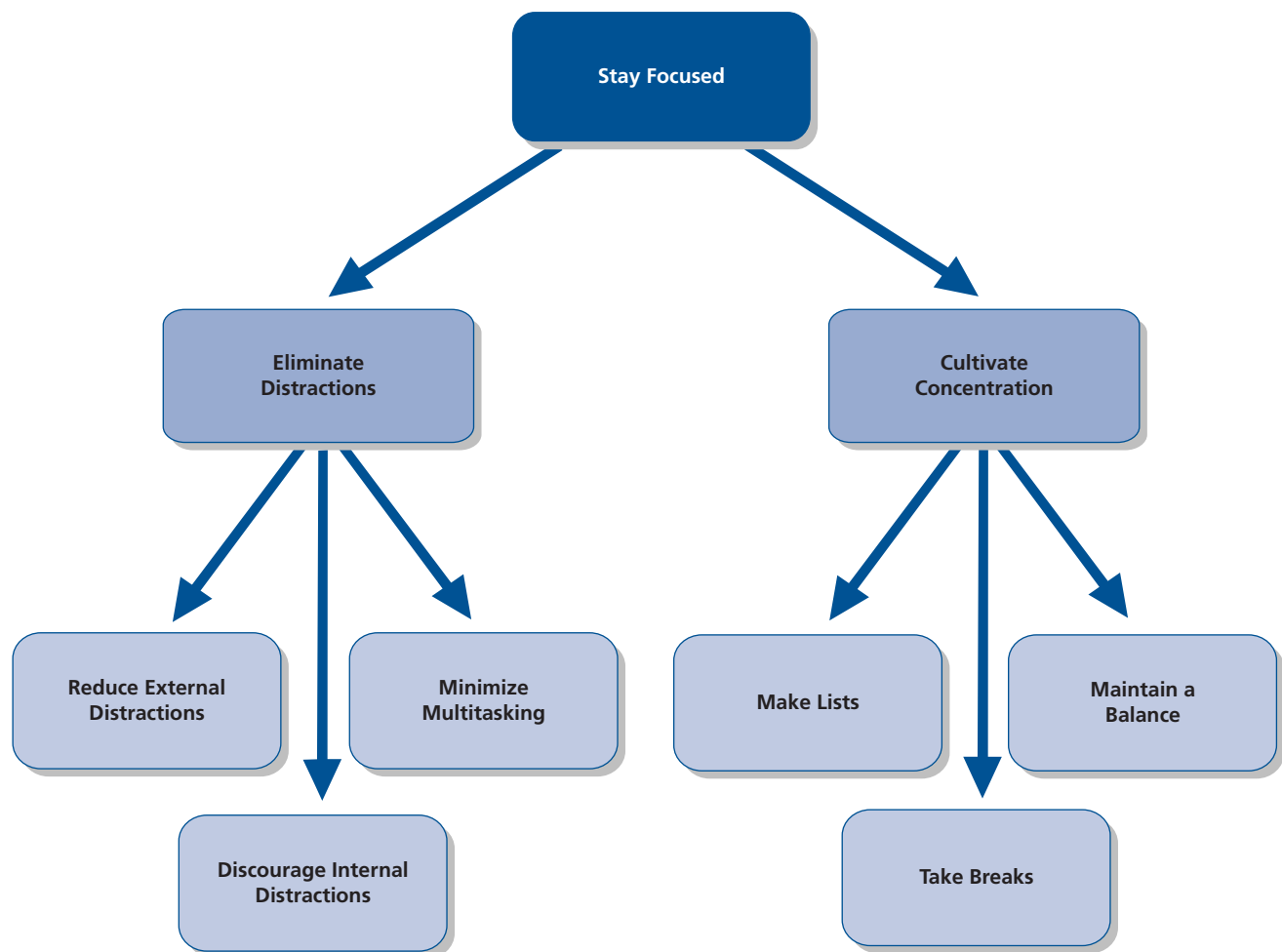
Deadline Facing a deadline can still be stressful. But once it meant risking your life.

deadline *děd' līn' n.* 1. A time limit, as for payment of a debt or completion of an assignment.
2. A boundary line in a prison that prisoners can cross only at the risk of being shot.*

These days we typically think of a deadline as a due date. In fact, that's been the word's predominant meaning since journalists adopted it in the early twentieth century. But where did the newspapers find the term in the first place? In prison. During the American Civil War, one of the most notorious prisons was Andersonville, a stockade in Sumter County, Georgia, where captured Union soldiers were detained under appalling conditions. The stockade itself was little more than a large plot of land, open to the elements but surrounded by a 15-foot wooden wall. Because Confederate guards feared their Union prisoners would try to vault the wall, they erected a second, smaller fence, roughly four feet tall and about three feet inside the stockade's main wall. This fence was known as the "dead-line." Any prisoner caught trying to cross it would be shot dead. Thanks to the country's vibrant newspaper business, which covered the Civil War extensively, Andersonville and its "dead-line" were soon infamous. Dead-line traveled from prison to printing press, where it was used to describe the border that each page of type had to fit within in order to be printed properly. If your type crossed the dead-line, those words wouldn't show up on the page. In time, the word went upstairs to the newsroom where editors turned "dead-line" into "deadline" and began to use it as a colorful term to describe a time limit that shouldn't be crossed by their reporters.**

* Adapted from "deadline." *The American Heritage Dictionary of the English Language*, 4th ed. Boston: Houghton Mifflin, 2000. <http://dictionary.reference.com/browse/deadline> (accessed May 2, 2009).

** Based on information from "deadline." *The American Heritage Dictionary of the English Language*, 4th ed. Boston: Houghton Mifflin, 2000. <http://dictionary.reference.com/browse/deadline> (accessed May 2, 2009).; "deadline." *Online Etymology Dictionary*. Douglas Harper, Historian. <http://www.etymonline.com/index.php?search=deadline> (accessed May 2, 2009).; "dead-line, *n.*" *Oxford English Dictionary*, 2nd ed. 20 vols. Oxford: Oxford University Press, 1989.; "dead-line, *n.*" *OED Online*. Oxford University Press. <http://sc-www2.santacruzpl.org:2062/cgi/entry/50058086> (accessed May 2, 2009).; "Andersonville Prison, Georgia, August 17, 1864 – Southwest view of stockade showing the dead-line" *Old Pictures*. 2006-2008. <http://www.old-picture.com/civil-war/Andersonville-Southwest-deadline-stockade.htm> (accessed May 2, 2009).



Concentrate all your thoughts upon the work at hand. The sun's rays do not burn until brought to a focus.

Alexander Graham Bell
(1847–1922), Scottish-born
Canadian scientist and inventor

Stay Focused

Everyone—from astronauts to athletes, from merchants to musicians—appreciates the value of being able to focus on the task at hand. Yet few of us know how to attain and then sustain this kind of concentration. Although concentration does not appear at the snap of the fingers, there are ways you can improve the conditions for concentration. To help you stay focused, no matter what your task may be, you need to:

- **Reduce External Distractions**
- **Overcome Internal Distractions**
- **Minimize Multitasking**
- **Cultivate Concentration**

What are some examples of concentration in sports?

What are some examples of concentration in academics?
How valuable is concentration?

What makes concentration elusive?

What does Figure 3.1 tell us about concentration?

Figure 3.1
A Goblet or Two Profiles?

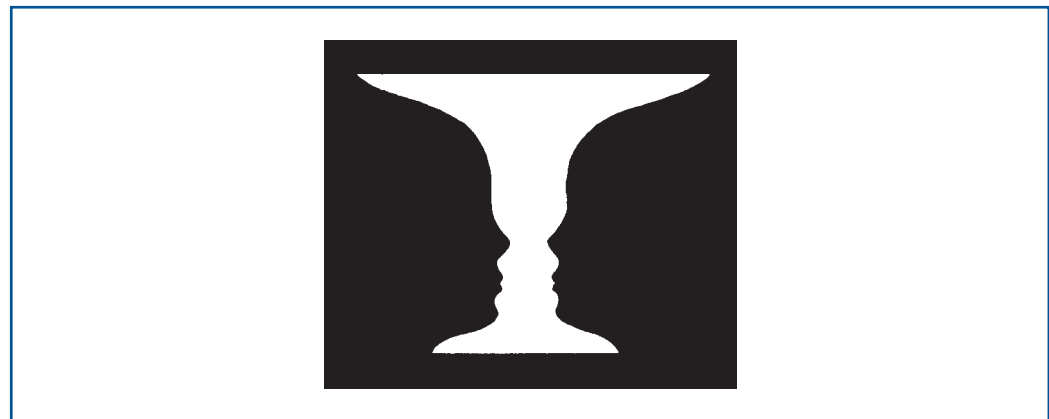
Concentration can sometimes be easier to describe than to define. Watch a good bowler as she takes her position and directs all her thoughts on knocking down the pins at the end of the alley. Watch a quarterback as he focuses on getting a pass to an open receiver, even while linebackers rush in on him from several directions. That's concentration! Or as world-famous tennis star Serena Williams once put it, "If you can keep playing tennis when somebody is shooting a gun down the street, that's concentration."

Of course, concentration isn't limited to sports. Have you ever become so absorbed in your textbook that you find yourself actually "talking" to the author? Have you ever been studying your notes so intently that when you finally look up, you realize that hours have passed since you last checked the clock? That's concentration!

Although concentration may be difficult to define, its value is crystal clear. American poet and essayist Ralph Waldo Emerson wrote that "Concentration is secret of strength," while management consultant Peter Drucker singled it out as "the key to economic results." Novelist Kurt Vonnegut called total concentration "the secret to success in any human endeavor," while legendary book editor Michael Korda described it as "the magic key that opens the door to accomplishment."

Concentration is focused thinking. Just as light waves can be focused into a single powerful beam—a laser—concentration can focus the power of your thoughts, enabling you to think with greater precision and penetrate difficult ideas. But powerful as it can be, concentration has an elusive quality. In fact, concentration comes only when you don't think about it. Ironically, if you were thinking deeply about a subject and suddenly realized that you were concentrating, at that moment you would have broken your concentration. Such prolonged, undivided attention can be difficult to achieve. After all, in normal circumstances there are dozens of things competing for your attention. Thoughts and ideas constantly bang, rattle, and knock on the door of your consciousness, trying to gain entry.

Figure 3.1 provides a vivid illustration of the natural tendency to divide attention. As you gaze at this picture, you'll probably discover that your visual focus is shifting



every few seconds so that you first see a goblet, then two profiles, and then the goblet again. Once you're aware of both images, it is difficult for your eyes to focus on one and ignore the other. Similarly, it is hard for your mind to focus on one idea at a time. People who can focus exclusively on the task before them have a much better chance of completing that task more quickly and accurately than those who divide their attention even when they don't mean to do so.

How can you concentrate if you can't do it directly?

So how do you stay focused? Because you can't strive for concentration directly, you must instead try to improve the conditions that promote concentration. That involves eliminating distractions and adopting strategies to cultivate concentration.

Eliminate Distractions

What's the first step in eliminating distractions?

Trouble concentrating may come from external distractions, such as sights or sounds that compete for your attention; from internal distractions, such as worries or daydreams; and from multitasking, which can splinter your concentration into unproductive pieces. Once recognized, these obstacles to concentration can be overcome.

Reduce External Distractions

FOCUSING

What is an external distraction?

Anything that stimulates your senses and in the process disrupts your concentration can be considered an external distraction. Study halls and living quarters are overflowing with such distractions, with everything from banging doors to baking bread. To work in a way that is compatible with concentration, you need the proper environment as well as the right equipment.

Select the Proper Environment

What elements make up a proper study environment?

Your study environment should be used exclusively for study. In addition, it should be quiet and well lighted. You can take your work to a place that is already designed for study, or you can create your own study environment.

What is the best environment for studying?

Find a Workshop You'd be hard-pressed to find an environment more suitable for high-quality concentration than a library. It offers a minimum of nonacademic distractions, a quiet atmosphere (usually as a matter of policy), and sufficient lighting. Get in the habit of studying at the library right away, on the first day of class. Even the walk to the library can be used productively as a review session or a refreshing break.

Why is it important to use your study area for studying only?

Whether you choose to work in the library or somewhere else, make sure your study area is reserved for studying. Psychologists emphasize that a conditioning effect is created between you and the desk: If you nap or daydream a lot while sitting at the desk, the desk can act as a cue for napping or daydreaming. By the same token, if you read or

What other steps can you take to ensure a proper study environment?

IT'S YOUR Q

What can you do to eliminate noise when you study?

How does music affect concentration?

What can you do to prevent e-mail and the Internet from becoming distractions?

work in bed, you make it difficult to work energetically and fall asleep easily. To avoid this negative conditioning, use your desk only for studying. When you feel the urge to nap or daydream, leave your desk to nap or daydream elsewhere.

Your study area should be your workshop, a place where you feel secure and comfortable. You can ensure that you have the proper environment for study and concentration if you minimize visual distractions, avoid or eliminate distracting noises, refrain from playing music, tune out text messages and the Internet, control the impulse to be disrupted by distractions, and provide the area with plenty of light.

Minimize Visual Distractions A sheet of notes or a page from a textbook can seem dull compared with a glimpse of a softball game being played on a nearby diamond or a view of gently falling snow. To improve your chances of concentration, avoid competition for your eyes' attention. Study by a window so you can take advantage of the natural light, but keep your head turned away from the potentially distracting view. Of course, not all visual distractions lie on the other side of the windowpane. If your study area contains photographs you're liable to look at, gadgets you're likely to fiddle with, or books you'll be tempted to pick up and read, remove them until you have completed your work.

Eliminate Noise If you need a quiet spot for efficient study, do your utmost to find one. Noise can be one of the most serious obstacles to effective study. Nothing is more wasteful than going over the same paragraph again and again because the noise makes it impossible to absorb what you are reading. If the library is the right place for you, make an effort to study there whenever you can. If you study at home, achieving quiet can sometimes be as simple as closing a door or inserting earplugs.

Turn Down the Loud Music When studying, loud music, especially vocal music, can break your concentration. To keep your concentration from bouncing between the music and your books, you expend energy that could be put to better use. However, some students can tolerate music better than others. If you find music nonintrusive, then instrumental music—played softly—could potentially form a pleasing background, and such a background can actually muffle some external, intermittent noises.

Tune Out Messaging and the Internet Avoid the temptation to surf the Web or constantly check messages by keeping your screen out of your line of sight and your phone out of range. If you are working on your computer, log out of e-mail and limit your Web surfing to the work at hand. If you're expecting important e-mail, set a timer for an hour or so, and work without interruption before checking. When the timer goes off, check for mail, then start the timer again. Do the same for cell phone text messages. Or simply let them wait until after you're done.

What does the spider technique involve?

Try the Spider Technique A vibrating tuning fork held close to a spider's web sets up vibrations in the web. After the spider makes a few hurried investigations and finds no fly in the web, it learns to ignore the vibrations. The next time you are studying in the library and the door opens, don't look up. Controlling your impulse to look up will disturb your concentration the first few times. But very soon, like the spider, you'll learn to ignore these external disturbances.

What is the most important piece of concentration equipment?

Use the Right Equipment

Because proper lighting is an important component of your study environment, the right light should head your list of equipment to promote concentration and reduce external distractions.

What is the best kind of light for studying?

Find the Right Light Whether it comes from incandescent light bulbs, fluorescent tubes, or compact fluorescent bulbs, the best light for studying is bright, even, and steady (remember B, E, and ST).

Bright. The emission of light is measured in lumens. For studying, you need at least 2,500 lumens. Two incandescent 100-watt bulbs (1,750 lumens each) will meet that requirement. So will a double-tube fluorescent lamp or two compact fluorescent bulbs of approximately 25 to 30 watts each. Fluorescents cost more than standard bulbs, but they can be just as bright and can last up to 100 times longer!

Even. Shadows in your work area or "hot spots" caused by glare will tire your eyes and make concentration difficult. Get rid of glare by shielding your lamp with a shade and by using a light-colored, nonglossy blotter on your desk. Eliminate shadows by using two lamps, fluorescent light, or diffuse light.

Steady. A constant flicker will undermine concentration. If you use fluorescent light, compact fluorescent bulbs are less likely to strobe than traditional fluorescents. If you are using incandescent light, make sure the bulb is screwed in properly.

What are the drawbacks of studying under bad light?

Good lighting makes for good studying. By contrast, poor lighting causes eyestrain, general tension, headaches, and sleepiness—irritations that interfere with concentration. If you study under good light but your eyes still bother you, have them examined by an ophthalmologist or optometrist. Clear and comfortable vision is essential to good studying.

What is the pencil technique?

Use a Pencil to Catalyze Concentration A technique that does not fail is the simple, humble *pencil technique*. The technique is this: *Whenever you are working to learn, study with a pencil in hand.* And use it! For example, if you are reading a textbook chapter, stop after several paragraphs and very briefly, in your own words, write down the key points made by the author. If, after reading several paragraphs, you find that no words come to you,

you have no recourse but to go back and read the passage again. This time, read with determination and concentration to make sure you learn the key points. The secret: Activity promotes and almost ensures concentration. The pencil provides the activity!

What is the best kind of chair for promoting concentration?

Find the Right Kind of Chair More ink and more words have been wasted extolling the virtues of a straight-backed, hard-seated hickory chair than on any other single piece of study equipment. Forget it: Use a comfortable, well-cushioned chair. Staying awake or falling asleep does not depend on your chair; rather, it depends primarily on your method of study, your attitude and self-discipline, the light, the room temperature, and, of course, on how much sleep you've been getting. A hard, often uncomfortable straight-backed chair can't take the place of these basic requirements.

IT'S YOUR Q

Use a Bookstand An extremely practical piece of equipment is a bookstand—not a bookshelf or bookends, an actual stand placed on your desk that holds the book in a tilted position and holds the pages down so that they do not flip over. Older ones are made out of metal. Newer ones come in plastic. A bookstand can work for you in many ways. First, and very important, it can give you a feeling of readiness to study—a feeling of being a scholar in the traditional sense. This alone is worth many times the price of the stand. Second, the stand provides physical freedom. It eliminates the strain of continually holding the book open, pressing down on two sides to keep the pages from flipping over, tilting the book to avoid glare, and trying to find something heavy enough to hold the book open so you can free your hands to make notes. It permits you to sit back with arms folded, to contemplate and reflect on the meaning of what you are reading.

What other equipment can be helpful for concentration?

Keep Other Equipment Nearby Remember the mnemonic for intelligently organizing your workspace: CIA (Current, Imminent, and Archived; see Chapter 2). Make sure your current materials include any basic equipment that will help you to study without interruption. Depending on the classes you are taking and your personal preferences, that may mean a dictionary, a calculator, a clock, a calendar, paper, notebooks, paper clips, tape, rubber bands, pencils, pens, erasers, and index cards. If you make it a habit to keep your desk well stocked, you won't derail your concentration with unplanned emergency trips to obtain necessities.

Discourage Internal Distractions

FOCUSING

SELF-DIRECTION

What are internal distractions?

Internal distractions are distractions that *you* create: daydreams, personal problems, anxiety, indecision, forgetfulness, and unrealistic goals. These distractions are as disruptive as the sights, sounds, and smells that make up the external variety, even though in this case the only one aware of them is you. Because internal distractions come from within, you have the power to eliminate or at least control them.

What is a concentration scoresheet?

Use a Concentration Scoresheet

Keep a sheet of paper handy. Whenever you catch your mind wandering, keep score by putting a check mark on the sheet. The mere act of doing this reminds you to get back to work. Students report that when they first tried this system, they accumulated as many as twenty check marks per textbook page, but after one or two weeks, they were down to one or two check marks per page.

How does keeping score help you concentrate?

The concentration scoresheet encourages self-observation. Making note of your breaks in concentration—when they happen, how often they occur, and what triggers them—will help you realize just how intrusive the lapses are and will enable you to gradually eliminate them.

What is the purpose of a worry pad?

Put Stray Thoughts on a Worry Pad

Although pleasant plans and diverting daydreams can be major sources of internal distraction, nagging worries and obligations can also take your mind off your work. The concentration scoresheet will alert you to these breaks in your attention, but it won't address the problems that prompted the distractions. To prevent the same worries from interfering with your concentration again and again, you must address them. A worry pad provides an excellent short-term solution to the problem.

How do you use a worry pad?

When an intrusive thought disrupts your concentration, write it down on your worry pad with the idea that you will attend to it just as soon as you get the chance. Then with your conscience clear and your bothersome thought recorded on paper, you can get back to the business of concentration. After you have finished studying, read over your list and give these concerns your full attention. If you cannot alleviate them yourself, get the help of friends or counselors.

What is the Stop! technique?

Use the Stop! Technique

Of course, not all distractions are worth writing down. Sometimes thoughts will flit into your mind that seem to have no other purpose than to throw you off course. The Stop! technique, suggested by authors Judith Greenbaum and Geraldine Markel in their book *Finding Your Focus*,¹ is a red light for distractions. It's designed to arrest any dark thoughts or daydreams before they have a chance to gain momentum.

How do you use the Stop! technique?

The Stop! technique is a simple and powerful tool that relies on what psychologists call self-talk. Whenever you find that you're heading in the wrong direction because of distractions or you feel you've been spinning your wheels, just tell yourself "Stop!" That's right. Say the word. Do it out loud and with authority if you can. Letting yourself know you're aware that your focus has been drifting and that you don't intend to let it go any further is often enough to set you right back on the road to concentration. Try it.

¹Judith Greenbaum and Geraldine Markel, *Finding Your Focus* (New York: McGraw-Hill, 2006), pp. 36–37.

Minimize Multitasking

FOCUSING

SELF-DIRECTION

IT'S YOUR Q

As we've seen, distractions are unwanted intrusions that vie for your attention and make concentration difficult if not impossible. Now imagine if you deliberately subjected yourself to several distractions at once. Sounds crazy, eh? That may be so, and yet many of us do it all the time. We call it multitasking. Although we might think of each concern as a task, not a distraction, when you're focusing on one task, all the others are distractions. For example, if you're doing research for a term paper, answering e-mail, listening to the radio, and finishing your lunch all at once or in rapid succession, you're almost certainly multitasking. *Multitasking* means working on several tasks simultaneously or rapidly shifting back and forth between them. The term comes from computer operating systems, which are able to keep several applications running at the same time. The difference, of course, is that people are not machines. Sometimes mistakenly held up as a model of efficiency, multitasking is not only less productive than working on one task at a time, it's also a troubling source of stress.² Luckily, there are a handful of simple steps you can take to avoid it.

Why does multitasking make you less productive?

Multitasking Makes You Less Productive

Dividing your brainpower between several tasks at once has a predictable effect. As a study published in the *Journal of Experimental Psychology* concludes, people who multitask are less efficient than people who don't. According to the study's coauthor, Dr. David Meyer, "There's scientific evidence that multitasking is extremely hard for somebody to do, and sometimes impossible."³

What are the costs of multitasking?

Of course, in spite of the evidence, thousands of people continue to multitask and would appear to be doing so successfully. "It doesn't mean you can't do several things at the same time," says Dr. Marcel Just, codirector of Carnegie-Mellon University's Center for Cognitive Brain Imaging. "But we're kidding ourselves if we think we can do so without cost."⁴ According to studies by Dr. Just, Dr. Meyer, and others, those costs include forgetfulness and difficulties in concentrating.⁵

²Christina Tibbits, "Just Throw 'Multitasking' Out Your Window of Work," *Austin Business Journal* (online), (June 10, 2002), available at <http://austin.bizjournals.com/austin/stories/2002/06/10/smallb3.html>.

³Sue Shellenbarger, "Juggling Too Many Tasks Could Make You Stupid," *Wall Street Journal Online* (February 28, 2003), available at <http://208.144.115.170/columnists/workfamily/20030228-workfamily.html>.

⁴Ibid.

⁵Melissa Healy, "We're All Multi-tasking, But What's the Cost?", *Los Angeles Times* (July 19, 2004), available at <http://articles.latimes.com/2004/jul/19/health/he-multitasking19>.

How does multitasking result in forgetfulness?

Forgetfulness. Chronic multitaskers often feel that their memories are failing them when in fact the converse is true: They are failing their memories. A 2006 UCLA study using functional magnetic resonance imaging (fMRI) confirmed this. It showed that attempting to learn while multitasking actually employs a different part of the brain called the striatum instead of the hippocampus, which is normally important for retaining facts and concepts. As a result, although the subjects of the study were able to sort index cards while multitasking, their ability to recall information from those cards was markedly reduced.⁶

What is it about multitasking that makes concentration difficult?

Difficulty in concentration. With multitasking, regardless of what you're working on, there's a good chance that at least part of your mind will be elsewhere. Nothing is as concentrated as it was initially when it's been diluted, and that includes the power of your thinking. Or, as Dr. Meyer explains, "No matter how hard you try, you will never be as good multitasking as you are concentrating on one [task]."⁷

Why does multitasking increase stress?

Multitasking Increases Stress

Not surprisingly, the drain on productivity results in an increase in the potential for stress.⁸ You are compelled to struggle against the inevitable forgetting that occurs and fight to maintain your concentration in a climate of relentless distractions. Moreover, multitasking forms an unhealthy alliance with two major culprits in stress—procrastination and loss of control—and can sometimes start you on a vicious cycle of insomnia as well. And some multitaskers have been exhibiting symptoms that are similar to those of attention deficit hyperactivity disorder.

What is the connection between multitasking and procrastination?

Procrastination. Another notorious source of self-imposed stress—procrastination—often works hand in hand with multitasking. (For more on procrastination and stress, see Chapter 4.) After all, one technique that procrastinators frequently use to put off assignments is to shift back and forth between several tasks instead of focusing on a single one. Avoid multitasking and you may succeed in discouraging procrastination in the same stroke.

Why does multitasking sometimes lead to a loss of control?

Loss of control. As we'll learn in Chapter 4, your sense of control can have a dramatic effect on your level of stress. Students, even overworked students, feel less stress when they are able to maintain a sense of control over what they are doing.

⁶Stewart Wolpert, "Don't Talk to a Friend While Reading This; Multi-Tasking Adversely Affects the Brain's Learning Systems, UCLA Scientists Report," available at <http://newsroom.ucla.edu/portal/ucla/Don-t-Talk-to-a-Friend-While-Reading-7212.aspx>.

⁷Amanda S. Fox, "Tallying the Cost of Doing Too Much," *CIO Magazine* (March 15, 2000).

⁸Larry Rosen and Michelle Weil, "Multitasking Madness," *Context* (Fall 1998), available at <http://www.contextmag.com/archives/199809/InnerGameOfWork.asp>.

Multitasking robs you of that control. All your activities and obligations are moving targets, and thus maintaining equilibrium becomes difficult for most and impossible for some.

IT'S YOUR Q

How does multitasking lead to attention-deficit trait (ADT)?

What are the symptoms of ADT?

How does ADT differ from attention-deficit hyperactivity disorder (ADHD)?

Insomnia. People who multitask—or attempt to—are often compelled to shift into a kind of mental overdrive that doesn't necessarily downshift when the day's work is done.⁹ A troubling side effect of multitasking, according to Drs. Larry Rosen and Michelle Weil, coauthors of the book *Technostress*, "is an inability to sustain a peaceful night's sleep or to enjoy what used to be calming, recreational times."¹⁰ As we'll see in Chapter 4, sleep loss not only results from stress, but can in turn be the source of *more* stress.

Multitasking Can Lead to Attention-Deficit Trait

Multitasking can short-circuit the sophisticated, reasonable area of your brain and hand control over to the ancient "fight or flight" region, an area fueled primarily by fear. As Dr. Edward Hallowell, an expert on attention-deficit trait (ADT) explains, "Fear shifts us into survival mode and thus prevents fluid learning and nuanced understanding."¹¹ The attempt to overload what are known as the brain's executive functions with too many tasks at once sends a distress signal to the primitive part of the brain, which abruptly takes over.¹² One minute you're a student, the next you're a cave man or cave woman, fleeing an unwelcome lion—all without even leaving your desk!

With ADT, your patience grows short, your thinking becomes black and white, your intelligence falters, and you are unable to effectively focus on the tasks at hand.¹³ In other words, for a student it spells disaster. According to Dr. Hallowell, "People with ADT have difficulty staying organized, setting priorities, and managing time."¹⁴

Students with ADT sometimes mistake their symptoms as being those of the attention deficit hyperactivity disorder (ADHD). Although people with ADT and those with ADHD share many of the same symptoms, there is a crucial difference. ADHD is a neurological disorder with a genetic component, whereas ADT springs entirely from your surroundings.¹⁵ If you have ADHD, you will probably always have ADHD. But with ADT, a change of scene and strategy can make the symptoms disappear. As Dr. Hallowell explains, "ADT can be controlled only by creatively engineering one's environment and one's emotional and physical health."¹⁶ One way to do this is by

⁹Ibid.

¹⁰Ibid

¹¹Edward M. Hallowell, "Overloaded Circuits: Why Smart People Underperform," *Harvard Business Review* (January 2005): p. 58.

¹²Ibid.

¹³Ibid.

¹⁴Ibid, p. 56.

¹⁵Ibid.

¹⁶Ibid, p. 59.

adopting the strategies for managing stress outlined in Chapter 4, which include eating well, exercising, and getting enough sleep. The other is to minimize multitasking.

Multitasking Can Be Minimized

What can you do to minimize multitasking?

Obviously, some multitasking is arguably beyond your control. If you're in the middle of an assignment and the phone rings or someone comes to your door, it's natural to want to answer. But most factors that lead to multitasking are well within the realm of your control. You can reduce the risk that multitasking will break your concentration by slowing down and planning your day carefully, by making a point to finish one task before beginning another, and by keeping a pad handy to deal with the distractions that may lead to multitasking in the first place.

How does slowing down help if you're feeling overloaded?

Given that many people multitask because they feel pressed for time, it might seem strange to suggest that slowing down can actually help improve both your attention and your productivity. But it can. If you move too quickly, you're apt to grow flustered and confused. If your executive functions get overloaded and succumb to your stone-age side, you're unlikely to get any quality studying done. On the other hand, if you take the time to figure out what's what, to listen carefully, ask questions, and think deeply, your tension should diminish and your attention should increase. You may be moving more deliberately, but you'll also be working more productively.

Why does a plan discourage multitasking?

Plan your day. Use the time-based scheduling outlined in Chapter 2 to inoculate yourself against multitasking. For many people multitasking provides a way of getting to work without ever having to decide which task is the most urgent, or the most interesting, or the most important. If each block of time has a specific task associated with it, you're less likely to bounce back and forth between several tasks.¹⁷

What can you do to make sure you work to completion?

Work to completion. Don't start a new task until you've finished the old one.¹⁸ This may be difficult with long-term tasks. But with short-term tasks, it should be achievable. If you do have a task that is stretched out over a longer period of time, try to create sub-tasks that you can tackle and complete before moving on.

What does a nearby pad do to prevent multitasking?

Keep a pad handy. The worry pad (page 63) is an excellent antidote to multitasking. It is designed to serve as a temporary holding area for pending ideas and concerns and can also be used to arrest early impulses to multitask. It's common to "task-shift" when there's a concern that "if I don't take care of it now, I never will." That's an understandable worry with a simple solution. Write it down, and get back to the task at hand. When you've finished the first job, you'll have the pad to remind you of what to do next.

¹⁷Rosen and Weil, "Multitasking Madness."

¹⁸Ibid.

Cultivate Concentration

What sorts of strategies will encourage concentration?

Although the best way to encourage concentration is usually to discourage distractions and keep multitasking to a minimum, you can also take positive actions to improve your concentration. Get into the habit of making lists, taking regular breaks, and maintaining a balance between the challenge of a particular assignment and your skill level.

Make Lists

PLANNING

What is the value of lists?

As we have seen, keeping random thoughts and information in your head instead of writing them down is a primary impediment to concentration. Lists allow you to free up your mind without losing important information. Use lists to remind yourself of day-to-day obligations and to catalog all the study equipment you're likely to need.

How can you prevent goals and appointments from distracting you? Are hand-held computers good tools for keeping lists?

To avoid worrying about the possibility of missing personal appointments and forgetting those things you've set out to do, write them down on your daily schedule (see Figure 2.4, page 36). This should enable you to shift smoothly from one activity to the next without breaking your concentration.

Some students may prefer to keep their lists on hand-held personal digital assistants (PDAs) that can be easily stowed in a pocket or purse. If you have a PDA and are comfortable using it, fine. But if you don't have a PDA, don't feel that you are somehow at a disadvantage. In general, it's quicker and more convenient to jot your list of items down on an ordinary paper pad. PDAs excel when you've got information you want to keep around for an extended period of time, but they aren't usually as time-efficient for making short-term lists. The tiny typewriter keypads or the handwriting recognition software that some PDAs employ may be ingenious, but they aren't flawless, and they do take some practice to perfect. It's important not to let keeping your list become its own distraction. This can be a danger with a handwritten list, but it's an even greater risk with a PDA.

Take Breaks

FOCUSING

How important is it to take breaks?

Breaks can sometimes be as vital as the work itself. Although some students with short attention spans may be tempted to take a rest after only a brief time of working, others have the exact opposite problem. As tempting as it may be to put your nose to the grindstone and just keep working until all of your assignments are completed, this sort of strategy can often be a recipe for distraction and burnout.

What happens if you don't take breaks?

If you allow physical energy to build up unabated, your mind will race. If you keep repressing concerns that compete for your attention, those concerns will eventually triumph and scuttle your concentration. And if you persist in denying such a basic instinct as hunger, ultimately all you'll be able to think of is food.

What are the benefits of taking a break?

If, however, you take a few minutes to defuse these distractions, stand up and stretch, address a problem you've been avoiding, or grab a healthy snack to tide you over, you can return to your work ready to concentrate.

How often should you take breaks?

Breaks are essential. But like almost anything beneficial, they're susceptible to abuse. If you're making progress, it's important not to take an open-ended break that interrupts your momentum. And if you're not making progress, don't let a break serve as an excuse for evasion. Be as precise about allocating your breaks as you are about scheduling your work time. In general, keep a ratio of no less than five to one between work time and break time. In other words, work for fifty minutes and then take a ten-minute break. Or work for twenty-five minutes and take a break of five minutes.

Maintain a Balance

SELF-DIRECTION

REFRAMING

How do you obtain the most rewarding kind of concentration?

Psychologist Mihaly Csikszentmihalyi believes that the most intense and rewarding kind of concentration (which he calls "flow") comes about when you develop a balance between the challenge of the work you are doing and the level of skills you possess.¹⁹ If the challenge of an assignment overwhelms your skill level, anxiety—not concentration—is likely to result. Conversely, if your skills are at a high level but the assignment isn't challenging, you're apt to become bored and easily distracted. Finally, if both your skill level and the challenge of an assignment are low, you'll probably grow apathetic and have no desire to concentrate (see Table 3.1).

How can you cultivate the balance of challenge and skills that lead to "flow"?

When your challenge and skill levels seem out of balance, there are several solutions to help get them in sync:

- *If your skill level is low*, you may need academic assistance. Your college should have a tutoring service. If not, see if you can get help from a classmate or even the instructor. If you recognize that you're having a problem, it's better to arrange for a tutor now instead of letting the problem fester. Occasionally, a perceived low skill level is just a matter of overly high expectations. Don't try to be an overnight success. Work methodically and patiently and there's a good chance that things will click. If after a time you still find you're floundering, then you may need a tutor's help.

¹⁹Richard Flaste, "The Power of Concentration," *The New York Times Magazine* (October 8, 1989), p. 26.

Table 3.1
Achieving Concentration by Balancing Challenge and Skills

Status	Symptom	Solution
Challenge and skill level low	Apathy	Group study; Tutoring
Skill level exceeds challenge	Boredom	Group study; Alternate text
Challenge exceeds skill level	Anxiety	Tutoring; Rethink goals
Challenge and skill level balanced	Concentration	No solution necessary: <i>Optimal condition for “flow”</i>

- *If your challenge level is low, you probably need some outside stimulus to boost your interest either in a class or for the text. Studying in groups can be a great way to liven up a course that doesn’t seem to challenge or excite you. The enthusiasm of other students can often prove infectious. An alternate solution is an alternate text. It may be that the voice, the style, or the organization of the required text doesn’t interest you. Reading another text on the same topic or using related workbooks or programmed materials that encourage active participation may boost your interest and put you in a better frame of mind for giving your course text another try. (For more on group study and alternate texts, see Chapter 8.)*

FINAL WORDS

How is learning to concentrate like gardening?

Ultimately, learning to concentrate is like learning to be a good gardener. Anyone who has had experience raising flowers, fruits, or vegetables knows that you can’t actually make them grow. The plant takes care of that, not you. All you can do is improve the conditions for growth. That requires skill, planning, and a little bit of luck. It’s the same with concentration.

CHAPTER CHECKUP

SENTENCE COMPLETION

Complete the following sentences with one of the three words or phrases listed below each sentence.

1. Concentration is focused _____.
light **thinking** **intensity**
2. The best light for studying is bright, even, and _____.
strong **stylish** **steady**
3. It's important not to let your efforts at keeping a list become their own _____.
distraction **advantage** **list**

MATCHING

In each blank space in the left column, write the letter preceding the phrase in the right column that matches the left item best.

- | | |
|-----------------------|---|
| _____ 1. "Flow" | a. Self-talk technique that discourages distractions |
| _____ 2. Library | b. Tires the eyes and makes concentration difficult |
| _____ 3. Stop! | c. Provides physical freedom and promotes readiness |
| _____ 4. Glare | d. Involves working on several things at once |
| _____ 5. Bookstand | e. Temporary measure for dealing with internal distractions |
| _____ 6. Multitasking | f. Helps you keep track of breaks in your concentration |
| _____ 7. Worry pad | g. Describes an especially rewarding kind of concentration |
| _____ 8. Check mark | h. Best environment for high-quality concentration |

TRUE-FALSE

Circle T beside the true statements and F beside the false statements.

1. T F Concentration only occurs when you're not thinking about it.
2. T F Anything that stimulates the senses is a potential external distraction.
3. T F E-mail and the Internet both have a positive influence on concentration.
4. T F Whether you stay awake or fall asleep depends on the type of chair you use.
5. T F Multitasking generally makes you more productive.

MULTIPLE CHOICE

Choose the word or phrase that completes each sentence most accurately, and circle the letter that precedes it.

1. Trouble concentrating is due primarily to
 - a. internal and external distractions.
 - b. boredom.
 - c. anxiety.
 - d. poor eyesight.
2. The face-goblet image in Figure 3.1 illustrates our natural tendency to
 - a. forget faces.
 - b. divide attention.
 - c. recognize people.
 - d. focus thoughts.
3. Although concentration is powerful, it is often
 - a. unnecessary.
 - b. elusive.
 - c. underestimated.
 - d. time consuming.
4. Multitasking can interfere with
 - a. sound sleep.
 - b. concentration.
 - c. consolidation.
 - d. all of the above.
5. When the challenge is high but your skill level is low, you will probably experience
 - a. concentration.
 - b. boredom.
 - c. anxiety.
 - d. apathy.

REFLECTION

Think about the ideas outlined in this chapter and then draw upon your own opinions and experiences to answer each question fully.

1. We don't all react in the same way to distractions. Which kinds of external distractions are most likely to break your concentration? What are some strategies you hope to use in the future to prevent them from causing you to lose your focus?

2. Of all the techniques mentioned in this chapter for encouraging concentration, which seems as though it will be most useful for you personally. Why?
3. Think of one of your more difficult classes, explain its current challenge/skill level relationship, and suggest actions you might take to bring things into better balance.

IT'S YOUR Q

The Q System uses marginal questions to encourage active reading. You'll notice that most but not all paragraphs in this chapter are accompanied by marginal questions. Now it's your Q. Scan the chapter for any paragraph that is missing a question, reread the paragraph, establish the main idea, and then arrive at a question that elicits it. Use the questions in the surrounding paragraphs as models for your own marginal questions.

VOCABULARY IN ACTION

To expand the horizons of your understanding and to refine the precision of your thought, the three exercises that follow are designed to help you grow, strengthen, and maintain your own vocabulary.

SAY WHAT?

From the three choices beside each numbered item, select the one that most nearly expresses the meaning of the italicized word in the quote. Make a light check mark (✓) next to your choice.

Any reviewer who expresses rage and *loathing* for a novel is *preposterous*. He or she is like a person who has put on full armor and attacked a hot fudge sundae.

—Kurt Vonnegut (1922–2007), American writer

- | | | | |
|------------------------|------------|----------|------------|
| 1. <i>loathing</i> | resentment | contempt | hatred |
| 2. <i>preposterous</i> | absurd | stupid | misleading |

Let's not be so *elitist* that we can't honor good, hard, dignified, *ennobling* work.

—Van Jones (1968–), American civil rights leader and environmentalist

- | | | | |
|---------------------|---------------|-----------|---------------|
| 3. <i>elitist</i> | condescending | wealthy | narrow-minded |
| 4. <i>ennobling</i> | royal | honorable | manual |

There is always an easy solution to every human problem—neat, *plausible* and wrong.

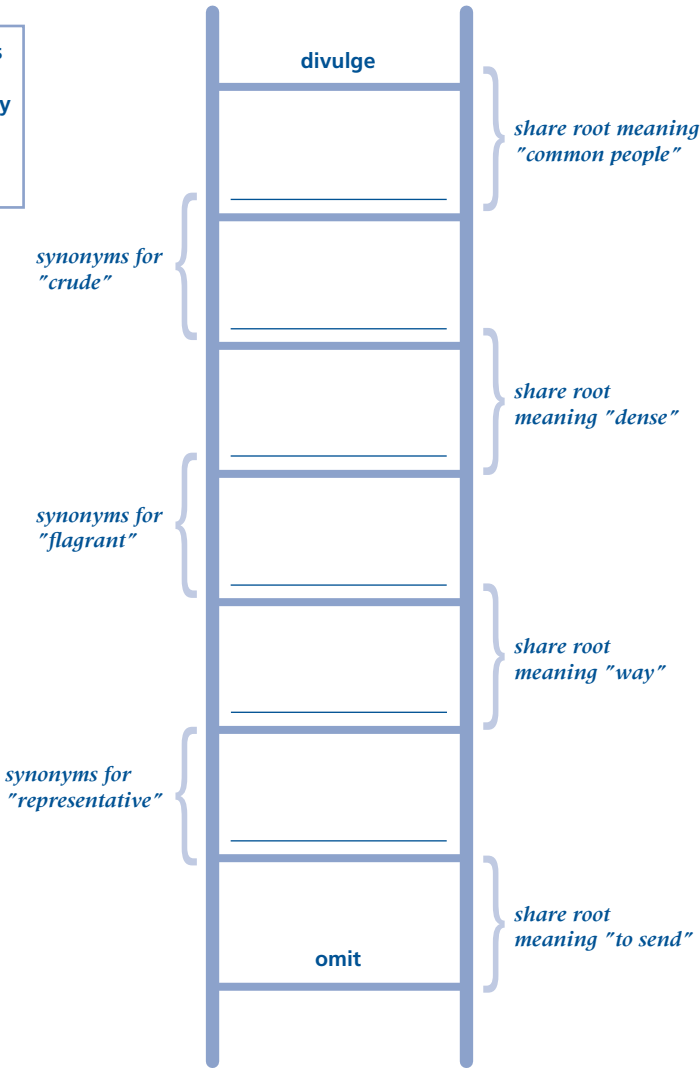
—H. L. Mencken (1880–1956), American editor and critic

- | | | | |
|---------------------|------------|-------|--------------|
| 5. <i>plausible</i> | reasonable | smart | advantageous |
|---------------------|------------|-------|--------------|

VOCAB-U-LADDER

Use your knowledge of word synonyms, prefixes, and roots to connect the word at the top rung to the word at the bottom rung, using the words listed below.

obvious
vulgar
emissary
gross
envoy
crass





Here's the story behind a word that figures prominently in the chapter you've just read.

Focus The warm fire behind our word for concentration

focus fŏ- kəs *n.* 1. A point in which the rays of light meet, after being reflected or refracted, and at which the image is formed. 2. A point so related to a conic section and certain straight line called the directrix that the ratio of the distance between any point of the curve and the focus to the distance of the same point from the directrix is constant. 3. A central point; a point of concentration.*

Four hundred years ago, “maintaining your focus” meant throwing another log on the fire. It’s puzzling even for some lexicographers to realize that our modern word *focus* came from a Latin word for hearth or fireplace. Even now, the transition is a bit of a mystery, although most experts acknowledge that it was German astronomer Johannes Kepler (1571–1630) who started the word *focus* on its modern path. Because he devoted his life—or you might say, his focus—to understanding the stars and the planets, Kepler was also interested in lenses and light, publishing a book called *The Optical Part of Astronomy* in 1604. It was in this book—now recognized as ushering in the modern study of optics—that Kepler first used the word *focus* to describe the point on a lens where light waves converge. Why did he give it that name? No one knows for certain, but as anyone who’s ever used the sun and a magnifying glass to burn a dry leaf knows, if you can bring together the strength of the sun’s rays at a single point, you can start a fire. According to the *Oxford English Dictionary*, it took almost another two hundred years for this term that dealt with light and waves to finally acquire its figurative meaning of “the center of activity, or area of greatest energy.” And that’s what we mean today when we use focus as a synonym for concentration.**

* Adapted from “focus.” *The American Heritage Dictionary of the English Language*, 4th ed. Boston: Houghton Mifflin, 2000. <http://dictionary.reference.com/browse/focus> (accessed May 1, 2009).

** Based on information from “focus.” *The American Heritage Dictionary of the English Language*, 4th ed. Boston: Houghton Mifflin, 2000. <http://dictionary.reference.com/browse/focus> (accessed May 1, 2009).; “focus.” *Online Etymology Dictionary*. Douglas Harper, Historian. <http://www.etymonline.com/index.php?search=focus> (accessed May 1, 2009).; “focus, *n.*” *Oxford English Dictionary*, 2nd ed. 20 vols. Oxford: Oxford University Press, 1989.; “focus, *n.*” *OED Online*. Oxford University Press. <http://sc-www2.santacruzpl.org:2062/cgi/entry/50087189> (accessed May 1, 2009).